



KENYATTA UNIVERSITY
INSTITUTE OF OPEN & VIRTUAL LEARNING
PLANT SCIENCES DEPARTMENT
SBT 318: TAXONOMY OF HIGHER PLANTS

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SBT 318: TAXONOMY OF HIGHER PLANTS

INTRODUCTION

Welcome to this module on the taxonomy of higher plants. This is a third year core unit in the Plant Sciences Department. You learnt the classification of the plant kingdom in your first year course on the Survey of the Plant Kingdom. The term taxonomy is derived from taxon which means a plant group; classification is about placing plants into groups and is the major part of plant taxonomy. In this module, I will start by defining the term taxonomy after which I will discuss the importance of it not just in conservation of plant biodiversity but also to other biological sciences. This is an interactive instructional module that will provide you with online learning experiences and activities.

The main purpose of this module is to expose you to the principles of plant taxonomy and the taxonomic processes. I will explain the rules of botanical nomenclature as they relate to the

units of classification and the naming of plants. I will discuss the nature and sources of characters which are the tools of the Plant Taxonomists in identification and classification. The collection, preparation and preservation of good quality herbarium specimens will be explained. I will also highlight the functions of the herbarium library and botanical gardens which are important accompaniments of the herbarium. Plant taxonomy relies a lot on floral characters for identification and classification, it is therefore important to learn the biology of these floral characters, this will be covered under reproductive data in plant taxonomy.

The skills learnt in this course will equip you with theoretical and technical skills to enable you to tackle taxonomic issues in your other biological courses, in future research projects, in collecting good quality plant specimens and in the identification of plant species.

TAXONOMY OF HIGHER PLANTS FLOW CHART

WEEK	TOPIC
WEEK 0	INTRODUCTION (YOUR CONTEXT, YOUR GOALS, THOUGHTS ABOUT TAXONOMY)
WEEK 1:	Definition and importance of plant taxonomy
WEEK 2:	Development of classification systems 1, Artificial

WEEK 3:	Development of classification systems 11: Natural
WEEK 4:	Development of classification systems 111 : Phylogenetic
WEEK 5:	Taxonomic units of classification
WEEK 6:	Characters and character states and their sources
WEEK 7:	Nomenclature
WEEK 8:	Identification and classification of cultivated plants.
WEEK 9:	Herbarium: Preparation of herbarium specimens and functions of Herbaria
WEEK 10:	Reproductive data in Plant Taxonomy
WEEK 11 & 12	End of semester Examinations

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OVERVIEW OF THE COURSE

Week 0: Introduction

This lesson is intended to help you acclimatize to the blended mode of learning. You will be interacting with other learners in the course thus creating a forum for you to motivate each other. You will be required to introduce yourself to the lecturer and your colleagues in the online or face to face sessions at the start of the course. It is also important to state your goals and context in this course.

Week 1: Definition and Economic Importance Plant Taxonomy

In this lesson, I will give you the definition of plant taxonomy and explain to you the importance of plant taxonomy. The purpose of this lesson is to prepare you for the content of the course and its benefits.

Week 2: Development of Classification Systems 1: Artificial

In this lesson I will explain the artificial systems of classification by: Theophrastus, Dioscorides, The Herbalists, Ceasalpinus. John ray, Tournefort and Linnaeus. The purpose of this lesson is to show the beginnings of formal classifications, the pioneers, the features they used and their contributions to plant taxonomy.

Week 3: Development of Classification Systems 11: Natural

In this lesson I will introduce you to the development of the natural systems of classification by Adanson, Bernard Jussieu, A. L Jussieu, De Candolle and Bentham and Hooker. The purpose of this lesson is to show you how taxonomic ideas progressed from the simple artificial systems to the more natural systems which grouped plants according to their overall resemblances. The lesson will highlight the authors of the natural systems and their contributions toward plant taxonomy.

Week 4: Development of Classification Systems 111: Phylogenetic

In this lesson, I will discuss the phylogenetic systems that were developed after Darwin's theories of evolution by German Botanists, Eichler, Engler and Prantl, American, Bessey, and British, Hutchinson. The purpose of this lesson is to show you how the various concepts on phylogenetic systems were developed over time, the botanists who developed them and their contributions to plant taxonomy.

Week 5: Taxonomic units of classification.

In this lesson, I will discuss the units of classification as prescribed by the botanical code of nomenclature; the hierarchical arrangements of the units will be explained, including the endings to names of some of the units. The sources of specific epithets will also be covered. The purpose of this lesson is to build on what the learners know about the units of classification and explain further the rules by the code regarding botanical names and derivation of specific epithets.

Week 6 Characters, character states and their sources

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In this lesson you will learn about taxonomic characters and character states or conditions and their sources. I will discuss the use of characters and the different types of characters. Taxonomic characters are derived from different sources such as morphology, anatomy, palynology, phytochemistry and cytology; the types of characters derived from these sources will be discussed. The purpose of this lesson is to equip learners with knowledge on the uses, types and sources of taxonomic characters.

Week 7: Nomenclature

In this lesson we explore the rules of the International code of nomenclature on the naming of plants. We will compare scientific and vernacular names, describe historical development of nomenclatural rules, and explain the rules on typification, principle of priority, conditions for valid publications, synonyms, author's names and rejection of names. The purpose of this lesson is to familiarize the learner with some of the rules of nomenclature regarding plant names.

Week 8: Identification and classification of cultivated plants.

In this lesson I introduce you to the methods of plant identification, the making of taxonomic keys, indented and bracketed keys. The challenges in the identification and classification of cultivated plants will be highlighted. The purpose of this lesson is to prepare you on the use of taxonomic keys for identification of plants and factors to note in the construction of taxonomic keys.

Week 9: Herbarium.

In this lesson, I will introduce you to the definition of a herbarium; explain the different types of herbaria, types of collections, describe how to collect good quality plant specimens. I will discuss what field notes to take, how to press and dry the specimens, mounting on herbarium sheets and preservation of specimens in the herbarium. I will describe the functions of the herbarium specimens, libraries and botanical gardens. The purpose of this lesson is to equip the learner with knowledge on what a herbarium is, its preparation, preservation and its importance to the modern world.

Week 10: Reproductive data in plant taxonomy,

In this lesson we look at the biology of the flower characters that are so much used for identification and classification. Discussion will be on types of reproductive data such as phenology, ultraviolet patterns, pollination mechanisms, abiotic and biotic, breeding systems and dispersal systems. The purpose of this lesson is mainly to highlight the biology behind the flower characters.

Week 11&12: Examination

Troubleshooting

If you have problems accessing or logging into the system (LMS):

Go to the Student Help Desk -<https://support.ku.ac.ke/>

Chat with us at <http://chat.ku.ac.ke/>

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Call the Digital school offices at (See our Contacts here) -<http://www.ku.ac.ke/dsvol/contact> us

Guides about how to use module are available at <http://www.ku.ac.ke/dsvol/students> resources

Also use our social media huddle

Twitter: @KUDigitalSchool

Facebook: @KUDSVOL

MODULE LEARNING OUTCOMES

By the end of the module the learner will be able to:

1. Describe the development of classification systems and the various concepts used in plant taxonomy
2. State the uses of characters, character states and their sources
3. Describe the rules of botanical nomenclature as regards the units of classification, typification etc.
4. Identify unknown plant specimens using botanical Keys
5. Collect good quality plant specimens for taxonomic purposes and the herbarium
6. Evaluate the use of reproductive data as it relates to plant taxonomy

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COURSE DESCRIPTION

This is a core unit for third year students taking Bachelor of Education and Bachelor of Science degrees. The rationale of offering the course is to equip students with skills and knowledge to undertake other biological courses where the principles and practical skills of taxonomy are necessary such as in research projects. The students do not need any prerequisite unit although the Morphology and Anatomy course taken as a first year course is quite relevant for the course. Emphasis will be placed on the development of classification systems, units of classification, characters and their sources, nomenclature, herbarium techniques, reproductive data in plant taxonomy, identification and practical flower studies. The course will take 35 instructional hours, some of which will be online and others face to face.

COURSE REQUIREMENTS

This is a blended learning course that will utilize learning materials and instructions online and the lessons will be self-guided with the lecturer being available for online and for face to face sessions. You will actively participate in your search for knowledge by undertaking several online activities of which you will find instructions in this module. This means that some of the 35 instructional hours of the course will be delivered face to face while other lessons will be taught online through various learner and lecturer activities. It is important for you to note that one instructional hour is equivalent to two online hours. Three instructional hours will be needed per week. Out of these, one will be used for face to face contact with your lecturer (also referred as e-moderator in the online activities) while the other two instructional hours (translating to four online hours) will be used for online activities otherwise referred to as e-activities in the

lessons. This will add up to the 5 hours requirement per lesson .You are advised to follow the topic flow-chart given so that you cover at least a lesson every week.

You will be required to participate and interact online with your peers and the e-moderator who in this case is your lecturer. Guidelines for the online activities referred to as e-tivities will be provided whenever there is an e-tivity. Please note that since the online e-tivities are part of the learning process, they may be graded at the discretion of your e-moderator. Such grading will however be communicated in the e-tivity guidelines and feedback given as soon as possible after the e-tivity. The e-tivities will include but will not be limited to online

Assessment quizzes, assignments and discussions. There are also assessment questions that you can attempt at the end of every lesson to test your understanding of the lesson. The answers to all the assessment questions are at the end of the module after lesson 20. All the resources that have been used in this module in form of books and videos are available under the resources section after the answers to the assessment questions.

ASSESSMENT

The assessment will include continuous assessment tests (CATs) which will account for 30% of the marks. The CATs can be based on assignments, practicals or written CATs. The examination will account for 70% of the marks.

I wish you all the best success in this course.

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LESSON 1: DEFINITION, USE OF TERMS AND IMPORTANCE OF PLANT TAXONOMY**1.1 Introduction**

In this first lesson I will give a full definition of plant taxonomy and a few other general terms commonly used in the course. The definition covers the principles of taxonomy which will, in the later lessons be given detailed descriptions. The importance of plant taxonomy will be discussed in this lesson since it is important for other biological sciences for the correct names. It is of use especially to those students undertaking research projects as they have to collect good quality specimens, so they would require knowledge of the herbarium techniques. They would also need to know how to write botanical names correctly and be familiar with taxonomic principles. Plant taxonomy has no data of its own, its sources of data are from other disciplines such as cytology, morphology, palynology, phytochemistry and anatomy, yet other sciences depend on it for accurate names and classification systems. There are three main aims of plant taxonomy: one is to provide a convenient method of identification and communication.

Secondly is to provide a classification that shows the natural relationships of plants and lastly to interpret evolutionary relationships of groups of plants

1.2 Lesson Learning Outcomes

By the end of this lesson, you will be able to:

1.2.1 Define Plant Taxonomy and other terms in general use

1.2.2 Discuss the importance of Plant Taxonomy

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1.2.1 Definition of Plant Taxonomy and other terms

Plant taxonomy is a science that includes identification, nomenclature and classification of plants. It is often referred to as systematic botany.

Identification

Identification is the determination of a plant or group of plants (taxon) as being identical to another already known group. The identification may be arrived at by use of keys as in botanical books, revisions and Floras; illustrations and photographs or by comparison with already identified plant specimens in the herbarium. In some cases the plants may be new and hence need to be described and validly published. Identification is not synonymous with naming. No names need to be given in the process of identification for example; the unknown plant may belong to a group called species A, which has not been given a formal scientific name. The fact that the unknown plant has been recognized to belong to that group, it has been identified. In most cases, identification goes hand in hand with the correct name.

Nomenclature

Once the plant is identified, it is given its correct scientific name. Nomenclature is giving the correct scientific name for the plant and the rules that govern the application of those names e.g. the International Code of Nomenclature for Algae Fungi and Plants in the case of wild 11 plants (previously known as The International Code of Botanical Nomenclature ICBN)). This is a set of rules which guides all aspects of naming plants.

Classification

Classification is the placing of plants (or groups of plants) into groups and in accordance to the rules of nomenclature. Plant groups are classified into different taxa, for example into species, genera, families etc. Thus classification expresses inter-relationships between plants.

TYPES OF CLASSIFICATION

There are three types of classifications as follows:

Artificial classification

Artificial classification uses one or a few characters for the purpose of identification or classification for example, classification based on flower colour. Plants can have white, yellow, blue and red corollas, if plants are classified according to this criteria then one ends up with groups of plants whose members do not show natural affinity and related members are put in different groups. Artificial classification is therefore characterized by use of one or few characters depending on the purpose for classification. It can also be referred to as a special purpose classification. Carpenters use it when they select wood based on the colour, texture of the grain and hardness of wood. The early systems of classification were mainly artificial, based on a few characters for the purpose of identification. For example, Theophrastus grouped plants on the basis of habit that is trees, shrubs, undershrubs and herbs irrespective of other characters.

Natural or phenetic classification

Natural classification seeks to establish the natural relationships of plants based on overall resemblance, mainly on gross morphological characteristics. The system uses a correlation of as many characters as possible and the more characters used the more natural the classification becomes. The classification reflects the situation as it is in nature. The system may or may not reflect genetic and evolutionary history but in most cases it does. A natural classification has a predictive value that means characters not used in making a natural classification agree with the characters that make the natural group. In practice natural classification is the most used because phylogenetic evidence is not always available.

Phylogenetic classification

Phylogenetic classification is based on evolutionary and genetic relationships of plants. It puts emphasizes on monophyletic groups, that is a group whose members are descendants of a common ancestor. Only homologous characters are used, that is characters possessed by a group due to common ancestry. This is unlike natural classification where any characters of the plant can be used regardless of whether they are due to common ancestry or not. In phylogenetic classification many characters are used except that the characters have to be homologous. It can therefore be regarded as a modified form of natural system. Groups that have members with different ancestors are the result of convergent evolution and are said to be polyphyletic. In this classification it is important to distinguish analogous features from

12 homologous ones. Analogous features are those that resemble due to convergent evolution. That is two evolutionary lines resemble each other due to a response to environmental factors. The result is mutations that result to similar phenotypes. For example Cacti (Cactaceae) and succulent Euphorbias (Euphorbiaceae) have reduced leaves and succulent stems as a result of adaptations to xeric environments but not due to common ancestry. The post Darwinian classification systems were mainly phylogenetic showing the evolutionary relationships of plant groups expressed in form of phylogenetic trees.

Taxon (plural taxa) A taxonomic group at any rank,

Description: Stating the overall characteristics of a plant group especially the morphological ones

Diagnosis: stating the distinguishing characteristics of a group of plants

E-tivity 1.2.1 –Define taxonomy and types of classification

Numbering, pacing and sequencing	1.2.1
Title	
Purpose	To define Plant Taxonomy and types of classification
Brief summary of overall task	Watch video plant taxonomy defined watch video
Spark	
Individual task	Distinguish between the different types of classification. Answer in the discussion forum
Interaction begins	Comment on the answers give by your colleagues in the discussion forum
E-moderator interventions	1. Ensure that learners are focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress.

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	4. Round-up the e-tivity
Schedule and time	This task should take 1 hour

Next	Importance of plant taxonomy
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1.2.2. The Importance of Plant Taxonomy

1. Name, identify and classify plants into groups in a systematic manner

Plant Taxonomists provide other biological sciences with: a) A system of identifying plants, b) correct names and c) A classification which shows the natural relationship of plants. For example in forestry, every tree must be identified, named and classified. In agriculture and horticulture breeding experiments are done with named plants. The Plant Ecologist needs to identify, name plants and assess their relationships to the environment.

2. Collect information about plant species: their numbers, distribution, abundance and origins.

3. Conservation of plants

Indigenous forests are under pressure of destruction through human activities like cultivation, overgrazing, fires, selective felling of trees for timber and harvesting of other forest products like firewood and medicinal plants. Forests also suffer through effects of climate change, these results in disappearance of some plant species. It is the duty of a Plant Taxonomist to make checklists of plants in those areas with immediate danger of destruction, collect representative specimens and put them in the herbarium for preservation and advice Governments on the need for preservation of natural forests. The taxonomist also helps to save seeds for future breeding studies especially of wild relatives of economically important crops such as maize, wheat, rice, etc. The plants need to be studied for them to be conserved; Plant taxonomists therefore conduct research on forest plants as bases for conservation.

4. Descriptions

The Plant taxonomists provide descriptions of plants including those of potential food and medicinal value. Plants are described in botanical books; floristic works e.g. The Flora of Tropical East Africa in our area, revisions, and monographs or in other research works.

5. Phylogenetic studies

This involves the study of fossils. These are preserved plant fragments used in describing characters of the ancestral forms of plants. The study of fossils is done by plant taxonomists.

14 Importance of Plant Systematics

Botany No Comments

Plant systematic has great importance for the study of botany:

1. It is used to describe different species. The description of each new species is preserved. It is used for comparison.
2. Plant systematic is used to name different plants. It sets rule for nomenclature. ,This nomenclature has eliminated the confusion among different botanists. Now knowledge of plants can be shared between botanists of different countries without problem of language and culture.
3. Plant systematic develops evolutionary relationship among the different groups of plants. It gives evolutionary trends among the plants.
4. Plant systematic provides basis kw the comparison of morphological, anatomical and cytological structures among different structures.

5.

6. Plant systematic also provides basis of genetics. Genetic analysis are performed on the basis of systematic,
7. Plant systematic has great importance in agriculture and herbal medicines. It provides us economic importance of different plants.

E-tivity -1.2.2 Importance of plant taxonomy

Numbering, pacing and sequencing	1.2.2
Title	
Purpose	To discuss the importance of plant taxonomy
Brief summary of overall task	. watch videos watch video Importance of plant taxonomy Read: Importance of systematics

Spark	
Individual task	Outline the importance of Plant Taxonomy to the modern world. Answer in discussion forum
Interaction begins	Comment on the answers given by your colleagues. Do this in the discussion forum
E-moderator interventions	1. Ensure that learners are focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress. 4. Round-up the e-tivity
Schedule and time	This task should take 1 hour
Next	Development of classification systems

1.3. Assessment Questions

1. The term taxonomy is a science that includes: Ethnobotany, genetics and ecology
T/F

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2. Which of the following statements is not true about plant taxonomy?
 - a) Provides a means of identification
 - b) It gives a classification about the relationships of plant groups
 - c) It explains the biosynthetic pathways about plants under study
 - d) It gives descriptions of plant groups
3. Which of the following statements is not true about the role of plant taxonomy in conservation?
 - a) Provides inventories of plants occurring in the forest

- b) Preserves representative specimens in the herbarium
 - c) Collects seeds of wild relatives of crop plants
 - d) Protects the forests from encroachment
4. Phylogenetic classification tries to establish ancestral relationships of plants T / F

1.4. References

Definitions of Plant taxonomy

1. <https://www.youtube.com/watch?v=7GqFbLNIJdxE>
2. <https://www.youtube.com/watch?v=k6LdsC64z3E>

Importance of Plant Taxonomy

3. <https://www.youtube.com/watch?v=mbjdgD2FXt8>
4. <https://www.youtube.com/watch?v=t5mZrqQJPUU>
<https://biologyboom.com/importance-of-plant-systematics/>

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LESSON 2: DEVELOPMENT OF CLASSIFICATION SYSTEMS - ARTIFICIAL

2.1. Introduction

In this lesson we will focus our attention to the history of classification systems in order to learn how the various concepts in plant taxonomy were developed over time. The earliest classification systems were artificial; these were followed by natural and finally phylogenetic. The artificial systems were based on a few characters mainly habit and uses. The early formal classification systems were started by the Greeks.

2.2 Lesson Learning Outcomes

By the end of this lesson, the learner will be able to:

2.2.1 Describe classifications based on habit and uses

2.2.2. Describe the sexual system by Linnaeus

2.2.1 Classification systems based on habit and uses

Theophrastus (c.370-287 B.C.)

A Greek pupil of Aristotle was the first botanist to write a classification system. His classification is written in *De Historia Plantarum*. (Inquiry into plants). He classified and described plants 480 different plants. He divided plants into trees, shrubs, under shrubs and herbs. These again he divided into cultivated and wild types. He was a good morphologist and was able to recognize many morphological differences between plants e.g. superior and inferior ovaries, determinate and indeterminate inflorescences, fused and free petals.

Dioscorides (1st Century AD)

He was a Greek medical doctor who was interested in the medicinal properties of plants. He wrote *De Materia Medica* in which he described 600 species. He classified plants according to their uses, for example: oils, perfumes, spices, cereals etc. Although his system was artificial he was able to identify some natural groups such as the Labiatae and Umbelliferae. His work was the most referred work until the 16th century (1500 years).

16th Century Herbalists.

A lot of botanical progress was made in the 16th century. This is referred to as the age of exploration when many new plants were collected. A lot of new books known as the Herbals were published. These were medically oriented books in which many plants were described and illustrated. Each Herbal contained accounts of about 1000 plants and therefore it was possible for a good taxonomist to memorize all the plants. Some of the important Herbals were written by Jerome Bock, Valerius Cordus and Brunfels among others. The Herbals also contained many superstitious stories, which led to the “doctrine of signatures “. According to this if a plant form resembled an organ of the human body, that plant would be used to treat diseases of that organ. Accordingly, Liverworts whose thallus resembles the human liver was used to treat liver diseases.

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Ceasalpinus (1519-1603)

He was an Italian botanist who wrote his classification in *De Plantis*. In this work he classified about 1500 species. He was the first botanist to use external morphological characteristics of the plant in his classification. He first divided plants on the basis of growth habit and then subdivided the groups on the basis of fruit and seed characters. His later divisions were based on floral and vegetative characters. Although his classification was artificial, he recognized

some natural families such as the Umbelliferae, Leguminosae, Cruciferae and Compositae. He prepared one of the oldest herbarium with a collection of 768 plant specimens.

John Ray (1628-1705)

He was an Englishman who contributed a lot to the foundation of modern taxonomy. According to Ray all parts of the plants should be used for classification. He noted that there are two types of variations in plants, genotypic by internal factors and phenotypic due to external environmental factors. According to him only genotypic variation is of taxonomic importance. He presented his classification in *Methodus Plantarum*, a three-volume work that described 1800 species. His initial division was based on growth habit. His further divisions were based on a large number of floral and vegetative characters. He divided flowering plants into dicotyledons and monocotyledons.

Joseph Pitton de Tournefort (1656-1708)

A French botanist regarded as the founder of the concept of modern genera. He described many genera separately from the species. He wrote *Institutiones Rei Herbariae* where he classified 698 genera. He divided genera into two groups, those which are identified by the flower and fruit characters e.g. Ranunculaceae and those distinguished by vegetative features e.g. *Pinus*, *Larix* and *Abies* which differ in position of the leaves. Some of his generic names are still used today, such as *Quercus*, *Ulmus* and *Fagus*.

E-tivity 2.2.1 Classifications based on habit and uses

Numbering, pacing and sequencing	2.2.1
Title	Classification system based on habit and uses
Purpose	To describe artificial classifications based on habit and use
Brief summary of overall task	Read types-of-classifications

Spark	Basis of some artificial classification system <table><tr><th>Sl. No</th><th>Name</th><th>Classification parameter</th></tr><tr><td>01</td><td>Vedas 200 BC</td><td>Religious importance, use, medicine, flowering and fruiting. Etc.</td></tr><tr><td>02</td><td>Theophrastus ca. 300 B.C.</td><td>Size and life time (herbs, shrubs and trees)</td></tr><tr><td>03</td><td>Herbalists 1470-1670</td><td>Medicinal values</td></tr><tr><td>04</td><td>Tournefort 1700</td><td>Flower and fruit giving rise (Innovator of genus. Distinction between genus and species.)</td></tr><tr><td>05</td><td>Linnaeus 1735</td><td>Sexual behavior of plants (number, length and union of stamens and carpels)</td></tr></table>	Sl. No	Name	Classification parameter	01	Vedas 200 BC	Religious importance, use, medicine, flowering and fruiting. Etc.	02	Theophrastus ca. 300 B.C.	Size and life time (herbs, shrubs and trees)	03	Herbalists 1470-1670	Medicinal values	04	Tournefort 1700	Flower and fruit giving rise (Innovator of genus. Distinction between genus and species.)	05	Linnaeus 1735	Sexual behavior of plants (number, length and union of stamens and carpels)
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03	Herbalists 1470-1670	Medicinal values																	
04	Tournefort 1700	Flower and fruit giving rise (Innovator of genus. Distinction between genus and species.)																	
05	Linnaeus 1735	Sexual behavior of plants (number, length and union of stamens and carpels)																	
Individual task	Outline the contributions to classification systems by botanists before Linnaeus. Answer this in the discussion forum																		
Interaction begins	Comment on the answers by your colleagues in the discussion forum																		
E-moderator interventions	1. Ensure that learners are focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress. 4. Round-up the e-tivity																		
Schedule and time	This task should take 1 hour																		
Next	Sexual classification by Linnaeus																		

2.2.2 Sexual system by Carolus Linnaeus

Carolus Linnaeus (1707-1778)

He was a Swedish botanist who is regarded as the father of modern botany. A lot has been written about his contributions to botany than any other biologist. From an early age as a student he was interested in the sexuality of plants. In his studies on the sexual habits of plants, he found out that the sexual organs of the flower, the stamens and pistils had very constant in plants. It was this realization that made him develop his “Sexual System” of classification. He wrote *Genera Plantarum* in 1737. In this work he described all the genera he recognized. While he was a professor of botany and medicine at Uppsala University

Sweden, he wrote *Species Plantarum* in 1753, in two volumes. He classified about 6000 species in 1000 genera. Linnaeus described the species under their respective genera. The

19 genera were classified into 24 classes based on the number and arrangement of the stamens thus he had classes Monandria with one stamen, Diandria, two stamens, Triandria with three stamens and Tetrandria with four stamens etc. and class Monadelphia with fused filaments. Plants that lack flowers were put in class of their own. The classes were grouped into orders based on the number of carpels for example order Monogynia had one style or sessile stigma (1carpel), order Digynia had two styles or two sessile stigmas (2 carpels) etc.

The Sexual System was very artificial but it allowed many species to be identified and filed, i.e. it brought order to the many plants collected and described by the 18th century.

Another of the Linnaeus’ contributions is that he started the consistent use of the binomial system of nomenclature in his *Species Plantarum* (1753). Each species had two names, a genus name and a specific epithet. Before the publication of the book there were no rules on how species were named, there was a mixture of monomials, binomials and polynomials (phrase names). For example Pepper mint, *Mentha piperata* had the phrase name “*Mentha floribus capitatas foliis lanceolatis serratis subpetiolatis.*” Phrase names were very difficult to remember and botanists were quick to take the advantage of the binomial system, which became the standard way of naming species. The *Species Plantarum*, 1753 was later adopted by modern botanists as the starting point of the binomial system of nomenclature. Another contribution by Linnaeus is that he started the construction of a precise Latin terminology to botanical names.

Linnaeus later realized that his sexual system was very artificial and in the sixth edition of *Genera Plantarum* (1764) he proposed 58 natural orders many of which are equivalent to modern day families such as Gramineae, Palmae, Compositae and Umbelliferae. Another major contribution by Linnaeus is that he trained many students in the university, many who became great botanists. He also had prepared a herbarium which is still in existence today and kept by the Linnaean Society of London

E-tivity 2.2.2 _Linnaeus sexual system

Numbering, pacing and sequencing	2.2.2
Title	

Purpose	To explain the sexual system and the contributions of Linnaeus to Plant taxonomy
Brief summary of overall task	Watch video sexual system Read Linnaeus sexual system

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Spark	
Individual task	List the contributions of Linnaeus to Plant Taxonomy Do this in discussion forum
Interaction begins	Comment on the answers given by your colleagues in the discussion forum
E-moderator interventions	1. Ensure that learners are focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress. 4. Round-up the e-tivity
Schedule and time	This task should take 1 hour
Next	Natural classification

2.3 Assessment Questions

1. Who used the following features in his classification, Trees, shrubs undershrubs and herbs?

- a) Dioscorides
- b) Linnaeus
- c) Theophrastus
- d) John Ray

2. “Doctrine of signatures” was a concept developed by:

- a) The 16th Century Herbalists
- b) Darwin
- c) 19th Century missionaries
- d) Linnaeus

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3. The artificial classification written by Dioscorides was written in a book called:

- a) *De Plantis*
- b) *De Historia Plantarum*
- c) *Methodus Plantarum*
- d) *De Materia Medica*

4. The starting point of binomial system of nomenclature is regarded as:

- a) *Genera Plantarum* 1737
- b) *Species Plantarum* 1753
- c) *Familles des Plantes* 1763
- d) *Theories Elementaire de la Botanique* 1813

2.4 References

Types of classification

1. <https://www.cliffsnotes.com/study-guides/biology/plant-biology/systematics/types-of-classifications>

2. Linnaeus sexual system

https://onlinelibrary.wiley.com/doi/10.1111/j.0107-055X.2007.00098_12.x#:~:text=Linnaeus'%20sexual%20system%20is%20an,and%20pistils%20in%20the%20flowers.

3. <https://www.youtube.com/watch?v=u1u01XLiOwE>

LESSON 3: DEVELOPMENT CLASSIFICATION SYSTEMS - NATURAL**3.1 Introduction**

In this lesson we focus on natural classification systems based on many characters. Artificial classification brought together groups of plants which did not show much similarity and many of the botanists who followed Linnaeus were dissatisfied with it. They therefore tried to devise groups, which reflected natural groups that are based on a correlation of many characters. Many of these 18th century botanists were French and Swiss.

3.2 Lesson Learning Outcomes

By the end of this lesson, you will be able to:

3.2.1 Describe natural classifications by Adanson, Jussieu and Decandolle

3.2.2 Describe natural classification by Bentham and Hooker

3.2.1 Natural Classification by Adanson, Bernard Jussieu, A.L. Jussieu, De Candolle.

Michel Adanson (1727-1806)

He rejected Linnaeus artificial system because it could not deal with the rich tropical flora of Senegal where he worked. He published *Familles des Plantes* in 1763 ten years after Linnaeus published *Species Plantarum*. He described groups of plants equivalent to modern day families and orders. According to him many characters should be used and should be given equal weight, the so-called empirical approach. This means that no character is more important than the other

Bernard de Jussieu (1699-1777)

He was in charge of plants at the Royal Botanical Garden in Paris and LA Tranon at Versailles. He rearranged the plants in the gardens based on as many characters as possible. He devised arrangements based on overall similarity. Unfortunately he did not publish his system.

His system was published by his nephew, Antoine Laurent de Jussieu (1748-1836) in the *Genera Plantarum*. He classified 100 orders, which are equivalent to modern day families. He used many characters in his classification. He divided flowering plants into monocotyledons and dicotyledons. Further divisions were based on the nature of the perianth, and ovary position (hypogynous, perigynous and epigynous).

Augustine Pyramus de Candolle (1778-1841).

A. P. De Candolle for the first time used the term plant taxonomy for the science of naming and classification of plants. He published *Theorie Elementaire de la Botanique* (1813). In this work he wrote several rules that should be followed in a natural system of classification. He later wrote a monumental work called *Prodromus Systematis Naturalis Regni Vegetabilis* (1823-1873), in seventeen volumes. The last ten were written after his death by other authors under the editorship of his son Alphonse de Candolle. His work covered all the dicotyledons

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(included in this were the Gymnosperms) but the work was not completed as the monocots and the cryptogams (non- seed plants) were not covered. A.P. De Candolle also published many monographs

E-tivity 3.2.1 Natural Classification by Adanson, de Jussieu and De Candolle

Numbering, pacing and sequencing	3.2.1
TitlPlante	
Purpose	The purpose of this lesson is to describe the natural classification systems
Brief summary of overall task	Read : natural classification Watch video: natural systems of classification

Spark	
Individual task	Discuss the contributions to classification systems by De Candolle. Do this in the discussion forum
Interaction begins	Comment on the answers given by your colleagues in the discussion forum
E-moderator interventions	1. Ensure that learners are focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress.
Schedule and time	This activity should take 1 hour
Next	Classification system by Bentham and Hooker

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3.2.2 Natural classification by Bentham and Hooker

George Bentham (1800-1884) and Joseph Dalton Hooker (1817-1911).

These were British scientists based at the Royal Botanical Gardens Kew, England. At the time they were working on the classification, J.D Hooker was the director of the Gardens and therefore they had the advantage of having both facilities of the gardens and the herbarium. They published *Genera Plantarum*, a three volume work in which all genera were accurately described using morphological and anatomical characters. They followed the Decadolle system but theirs was more improved. They separated the dicotyledons from the Gymnosperms while Decandolle had placed them in the same group. In their classification based on the seed plants, they had three groups: the dicotyledons, gymnosperms and monocotyledons. They have been criticized for placing gymnosperms between the dicotyledons and the monocotyledons. Dicotyledons were divided into the following three groups: polypetalae (had free petals), Gamopetalae (had fused petals) and Monochlamydae (without petals). Many British herbaria and those of the commonwealth use the Bentham and Hooker system in arranging their plant

specimens. By the time the *Genera Plantarum* was published (1762-1763) Darwin had published “Origin of Species” (1859) but Bentham and Hooker did not use phylogenetic criteria which was taken up by latter botanists.

1. Artificial system of classification

Carolus Linnaeus (1707 - 1778) was a great Swedish Botanist and said to be the “**Father of Taxonomy.**” He outlined an artificial system of classification in “*Species Plantarum*” in 1753, wherein he listed and described 7,300 species and arranged in 24 classes mostly on the basis of number, union (adhesion and cohesion), length, and distribution of stamens.



Figure 5.4:
Carolus Linnaeus

The classes were further subdivided on the basis of carpel characteristics into orders. Hence the system of classification is also known as **sexual system of classification.**

This system of classification, though artificial, was continued for more than 100 years after the death of Linnaeus, due to its simplicity and easy way of identification of plants.

However the system could not hold good due to the following reasons.

1. Totally unrelated plants were kept in a single group, whereas closely related plants were placed in widely separated groups. Example:

- a. Zingiberaceae of monocotyledons and Anacardiaceae of dicotyledonous were placed under the class **Monandria** since these possess single stamens.
- b. *Prunus* was classified along with *Cactus* because of the same number of stamens.

No attempts were made to classify plants based on either natural or phylogenetic relationships which exist among plant groups.

2. Natural system

Botanists who came after Linnaeus realized that no single character is more important than the other characters. Accordingly an approach to a natural system of classification sprouted in France. The first scheme of classification based on overall similarities was presented by **Antoine Laurent de Jussieu** in 1789.

24 classes recognized by Linnaeus in his *Species Plantarum* (1753) on the basis of stamens.

No	Classes	Characters
1	Monandria	stamen one
2	Diandria	stamens two
3	Triandria	stamens three
4	Tetrandria	stamens four
5	Pentandria	stamens five
6	Hexandria	stamens six
7	Heptandria	stamens seven
8	Octandria	stamens eight
9	Ennandria	stamens nine
10	Decandria	stamens ten
11	Dodecandria	stamens 11-19
12	Icosandria	stamens 20 or more, on the calyx
13	Polyandria	stamens 20 or more, on the receptacle
14	Didynamia	stamens didynamous; 2 short, 2 long
15	Tetradynamia	stamens tetradyna- mous; 4 long, 2 short
16	Monadelphica	stamens monadelphous; united in 1 group
17	Diadelphia	stamens diadelphous; united in 2 groups
18	Polyadelphia	stamens polyadelphous; united in 3 or more groups
19	Syngenesia	stamens syngenesious; united by anthers only
20	Gynandria	stamens united with the gynoecium
21	Monoecia	plants monoecious
22	Dioecia	plants dioecious
23	Polygamia	plants polygamous

Bentham and Hooker system of classification

Figure 5.5: George Bentham and J.D. Hooker.

A widely followed natural system of classification considered the best was proposed by two English botanists **George Bentham** (1800 - 1884) and **Joseph Dalton Hooker** (1817–1911). The classification was published in a three volume work as “*Genera Plantarum*” (1862–1883) describing 202 families and 7569 genera and 97, 205 species. In this system the seeded plants were classified into 3 major classes such as Dicotyledonae, Gymnospermae and Monocotyledonae.

Class I Dicotyledonae: Plants contain two cotyledons in their seed, leaves with reticulate venation, tap root system and tetramerous or pentamerous flowers come under this class. It includes three sub-classes – **Polypetalae**, **Gamopetalae** and **Monochlamydeae**.

Sub-class 1. Polypetalae: Plants with free petals and dichlamydeous flowers come under polypetalae. It is further divided into three series – **Thalamiflorae**, **Disciflorae** and **Calyciflorae**.

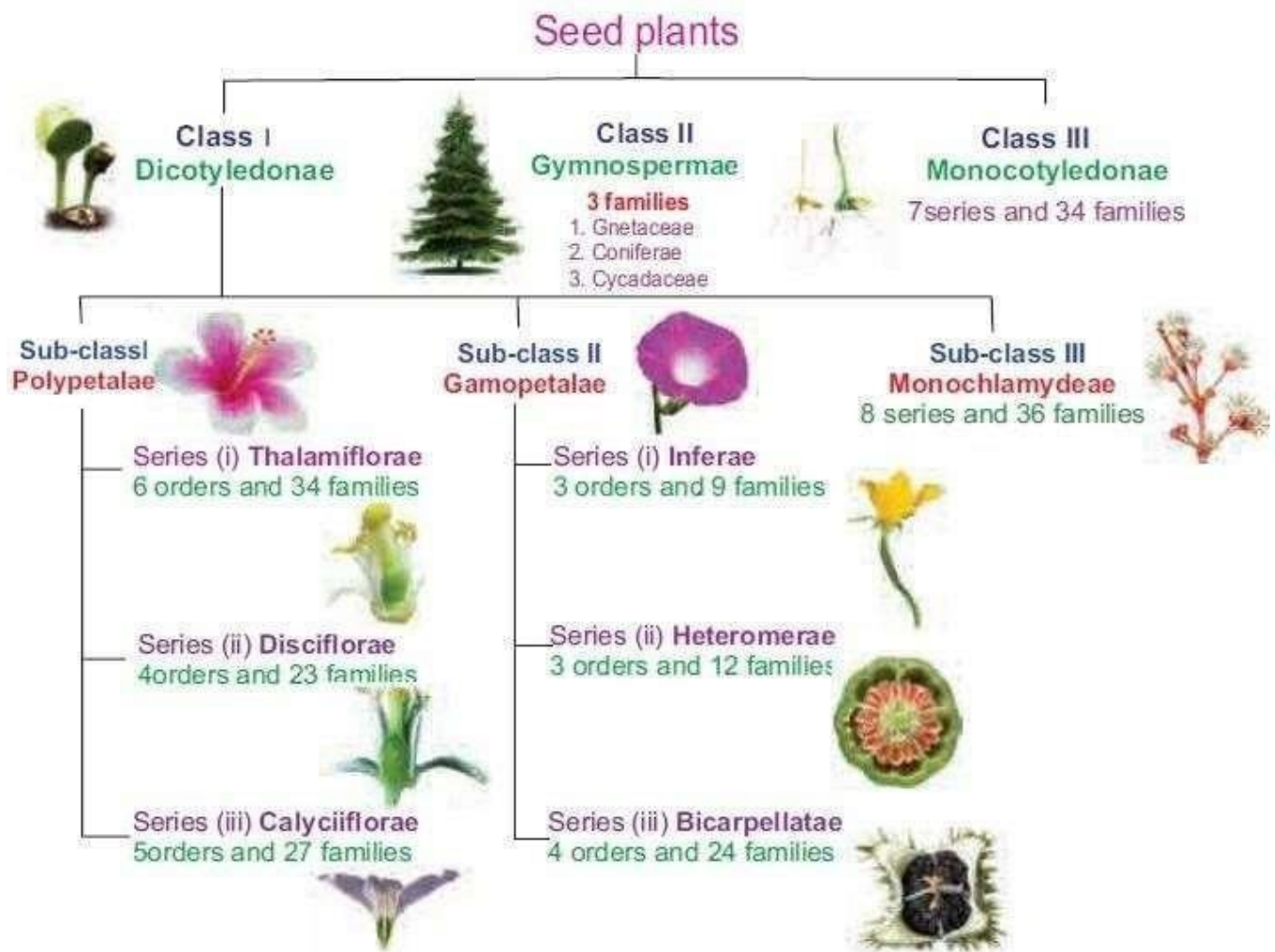


Figure 5.6: Bentham and Hooker system of classification

Series (i) Thalamiflorae: Plants having flowers with dome or conical shaped thalamus and superior ovary are included in this series. It includes 6 orders and 34 families.

Series (ii) Disciflorae: Flowers having prominent disc shaped thalamus with superior ovary come under this series. It includes 4 orders and 23 families.

Series (iii) Calyciflorae: It includes plants having flowers with cup shaped thalamus and with inferior or sometimes with half inferior ovary. Calyciflorae includes 5 orders and 27 families.

Sub-class 2. Gamopetalae: Plants with united petals, which are either partially or completely fused to one another and dichlamydeous are placed under Gamopetalae. It is further divided into three series – Inferae, Heteromerae and Bicarpellatae.

Series (i) Inferae: The flowers are epigynous and with inferior ovary. Inferae includes 3 orders and 9 families

Series (ii) Heteromerae: The flowers are hypogynous, superior ovary and with more than two carpels. Heteromerae includes 3 orders and 12 families.

Series (iii) Bicarpellatae: The flowers are hypogynous, superior ovary and with two carpels. Bicarpellate includes 4 orders and 24 families.

Sub-class 3. Monochlamydeae: Plants with incomplete flowers either apetalous or with undifferentiated calyx and corolla are placed under Monochlamydeae. The sepals and petals are not distinguished and they are called **perianth**. Sometimes both the whorls are absent. Monochlamydeae includes 8 series and 36 families.

Class II Gymnospermae: Plants that contain naked seeds come under this class. Gymnospermae includes three families – Gnetaceae, Coniferae and Cycadaceae.

Class III Monocotyledonae: Plants contain only one cotyledon in their seed, leaves with parallel venation, fibrous root system and trimerous flowers come under this class. The Monocotyledonae has 7 series and 34 families.

The Bentham and Hooker system of classification is still supposed to be the best system of classification. It has been widely practiced in colonial countries and herbaria of those countries were organized based on this system and is still used as a key for the identification of plants in some herbaria of the world due to the following reasons:

- Description of plants is quite accurate and reliable, because it is mainly based on personal studies from actual specimens and not mere comparisons of known **facts**.

- As it is easy to follow, it is used as a key for the identification of plants in several herbaria of the world.

Though it is a natural system, this system was not intended to be phylogenetic.

3. Phylogenetic system of classification

The publication of the *Origin of Species* (1859) by **Charles Darwin** has given stimulus for the emergence of phylogenetic system of classification.

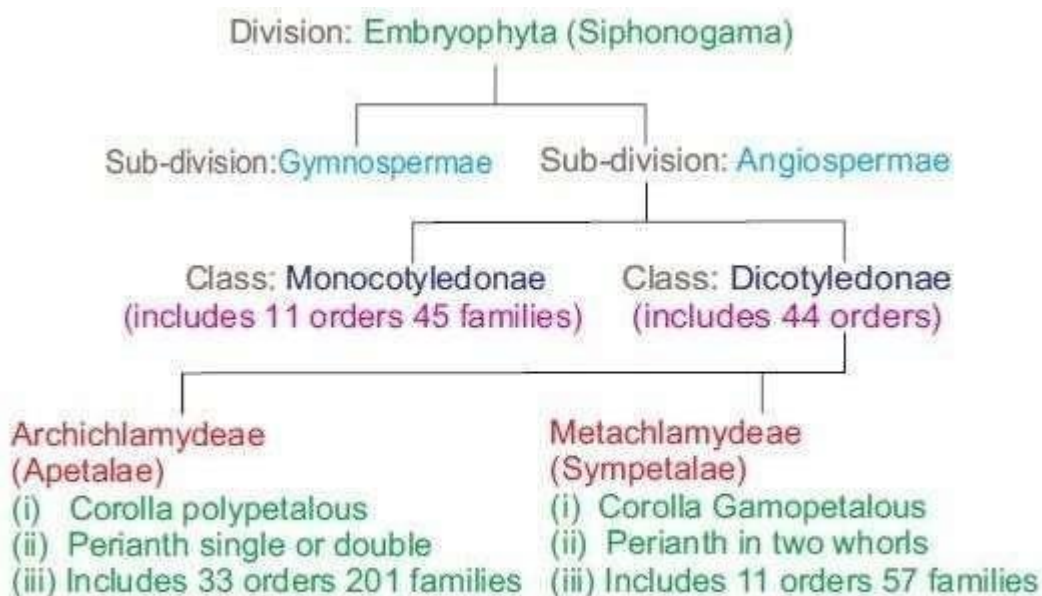


Figure 5.7: Outline of Engler and Prantl classification

I Adolph Engler and Karl A Prantl system of classification

One of the earliest phylogenetic system of classification of the entire plant Kingdom was jointly proposed by two German botanists **Adolph Engler** (1844 - 1930) and **Karl A**

Prantl (1849 - 1893). They published their classification in a monumental work “*Die Naturelichen Pflanzen Familien*” in 23 volumes (1887- 1915)



Figure 5.8: Adolph Engler and Karl A Prantl

In this system of classification the plant kingdom was divided into 13 divisions. The first 11 divisions are Thallophytes, twelfth division is **Embryophyta Asiphonogama** (plants with embryos but no pollen tubes; Bryophytes and Pteridophytes) and the thirteenth division is **Embryophyta Siphonogama** (plants with embryos and pollen tubes) which includes seed plants.

II Arthur Cronquist system of classification

Arthur Cronquist (1919 - 1992) an eminent American taxonomist proposed phylogenetic classification of flowering plants based on a wide range of taxonomic characters including anatomical and phytochemical characters of phylogenetic importance.

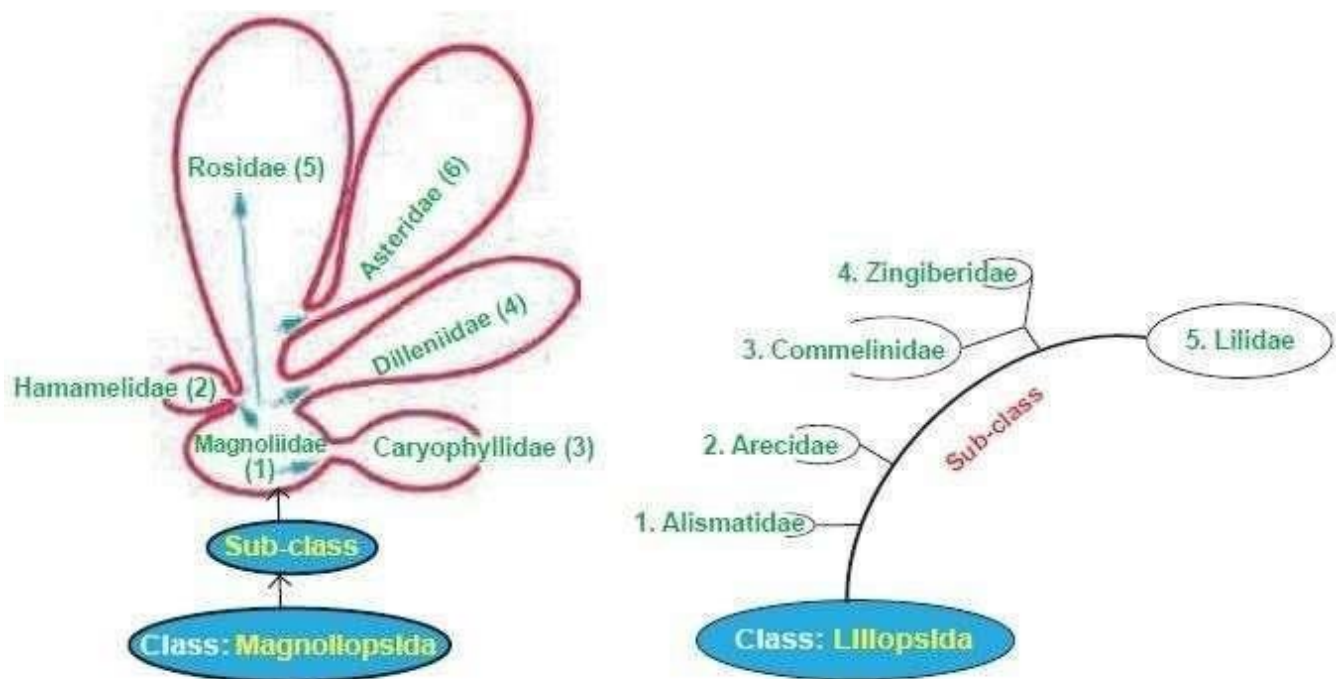


Figure 5.10: Diagrammatic representation of the relationship between class Magnoliopsida and Liliopsida.

He has presented his classification in 1968 in his book titled “**The evolution and classification of flowering plants.**” His classification is broadly based on the Principles of phylogeny that finds acceptance with major contemporary authors.



Figure 5.9: Arthur Cronquist

Cronquist classified the angiosperms into two main classes **Magnoliopsida** (=dicotyledons) and **Liliopsida** (= monocotyledons). There are 6 subclasses, 64 orders, 320 families and about 165,000 species in Magnoliopsida, whereas in Liliopsida there are 5 sub classes, 19 orders, 66 families and about 50,000 species.

The Cronquist system of classification also could not persist for a long time because the system is not very useful for identification and cannot be adopted in herbaria due to its high phylogenetic nature.

4. Angiosperm phylogeny group (APG) classification

The most recent classification of flowering plants based on **phylogenetic data** was set in the last decade of twentieth century. Four versions of Angiosperm Phylogenetic Group classification (APG I, APG II, APG III & APG IV) have been published in 1998, 2003, 2009 and 2016 respectively. Each version supplants the previous version. Recognition of **monophyletic** group based on the information received from various disciplines such as gross morphology, anatomy, embryology, palynology, karyology, phytochemistry and more strongly on molecular data with respect to DNA sequences of two chloroplast genes (*atpB* and *rbcL*) and one nuclear gene (nuclear ribosomal 18s DNA).

The most recent updated version, APG IV (2016) recognised 64 orders and 416 families. Of these, 416 families 259 are represented in India.

The outline of APG IV classification is given below.

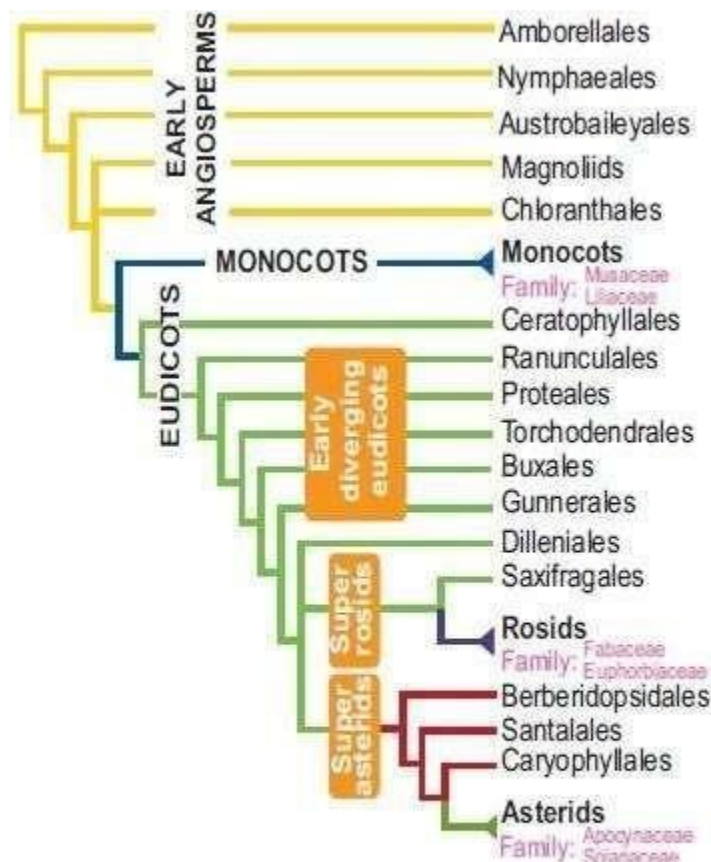


Figure 5.11: Simplified version of APG IV
(Source: Plant Gateway's The Global Flora,
Vol. I January 2018)

Angiosperms are classified into three clades early angiosperms, monocots and eudicots. Early angiosperms are classified into **8 orders** and 26 **families** (**ANA**-grade + magnoliids + Chloranthales)

Amborellales **N**ymphaeales **A**ustrobaileyales

- Seeds always with two cotyledons.
- Presence of ethereal oils.
- Leaves are always simple net-veined.
- Each floral whorls with many parts.
- Perianth usually spirally arranged or parts in threes.
- Stamens with broad filaments.
- Anthers tetrasporangiate.

- Pollen monosulcate.
- Nectaries are rare.
- Carpels usually free and.
- Embryo very small.

Monocots are classified into 11 orders and 77 families (basal monocots + lilioids + commelinids)

- Seeds with single cotyledon.
- Primary root short-lived.
- Single adaxial prophyll.
- Ethereal oils rarely present.
- Mostly herbaceous, absence of vascular cambium.
- Vascular bundles are scattered in the stem.
- Leaf simple with parallel-veined.
- Floral parts usually in threes.
- Perianth often composed of tepals.
- Pollen monosulcate.
- Styles normally hollow and.
- Successive microsporogenesis.

Eudicots are divided into 45 orders and 313 families (early diverging eudicots + super rosids + super asterids).

- Seeds with always two cotyledons.
- Nodes trilacunar with three leaf traces.
- Stomata anomocytic.
- Ethereal oils rarely present.

- Woody or herbaceous plants.
- Leaves simple or compound, usually net-veined.
- Flower parts mostly in twos, fours or fives.
- Microsporogenesis simultaneous.
- Style solid and .
- Pollen tricolpate.

APG system is an evolving system that might undergo change periodically based on the new sets of data from various disciplines of Botany. It is the currently accepted system across the world and followed by all the leading taxonomic institutions and practising taxonomists. However, it is yet to percolate into the Indian botanical curriculum.

Changes in earlier taxonomic understanding.

The newly proposed APG classification system has brought many changes in our earlier understanding on the concept of primitive flowering plant families. Some of them are given below:

- The real Ranalean families, especially the arborescent ones, are no more the primitive families. But as per APG classification system, Amborellaceae, Nymphaeaceae, Austrobaileyaceae, Magnoliaceae and Chloranthaceae form the early angiosperms.
- Monocots are recognised as a monophyletic group and hence terminology is retained.
- Dicots are polyphyletic group and as a result the use of the term dicotyledons as a group becomes outdated.
- Liliaceae (Sensu lato) is split into 14 families.

- Molluginaceae and Gisekiaceae are recognised separately from Aizoaceae.
- Euphorbiaceae (s.l.) is split into Phyllanthaceae, Picrodendraceae and Putranjivaceae.
- Asclepiadaceae are merged with Apocynaceae (s.l.)

Many genera that were conventionally treated under Verbenaceae such as *Clerodendron*, *Tectona* and *Vitex* are transferred to Lamiaceae based on the modified circumscription of these families.

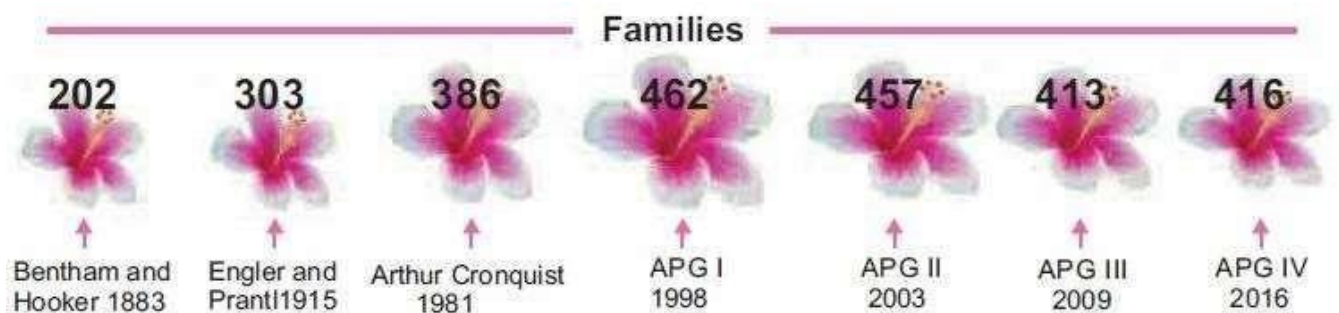


Figure 5.12: A timeline showing the history of classifying flowering plants into families.
(Source: Royal Botanic Gardens Kew State of World's Plant 2017)

Classification reflects the state of our knowledge at a given point of time. It will continue to change as we acquire new information.

A significant number of major herbaria, including Kew are changing the order of their collections in accordance with APG.

The influential world checklist of selected plant families (also from kew) is being updated to the APG III system.

A recent photographic survey of the plants of USA and Canada is organized according to the APG III system.

In UK, the latest edition of the standard flora of the British Isles written by Stace is based on the APG III system.

- [Prev Page](#)
- [Next Page](#)

E-tivity 3.2.2 Natural Classification by Bentham and Hooker

Numbering, pacing and sequencing	3.2.2
Title	
Purpose	The purpose of this lesson is to describe the natural classification systems
Brief summary of overall task	Read on classification system by Bentham and Hooker Essay on natural systems Watch video: natural systems of classification

Spark	<i>Outline of Bentham and Hooker's classification of plants</i>
Individual task	Outline the contributions to natural classification systems by Bentham and Hooker. How does it differ from the artificial system of Linnaeus? Answer this in the discussion forum
Interaction begins	Comment on the answers given by your colleagues in the discussion forum
E-moderator interventions	- Ensure that learners are focused on the contents and context of discussion. - Stimulate further learning and generation of new ideas. - Provide feedback on the learning progress.
Schedule and time	This activity should take 1 hour
Next	Phylogenetic classification systems

3.3. Assessment questions

1. Natural classification is based on:

- A few characteristics of the plant
- A correlation of many characteristics**
- Homologous characters only
- Anatomical characters only

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2. The first botanist to use the term "plant taxonomy" was:

- a) Dioscorides
- b) Linnaeus
- c) Caesalpinus
- d) **De Candolle**

3. The first rules on natural classification were written in a book called:

- a) ***Theory Elementaire de la Botanique***
- b) *Genera Plantarum*
- c) *Species Plantarum*
- d) *De Materia Medica*

4. Bentham and Hooker classified seed plants based on:

- a) Cytological data
- b) Chemical data
- c) **Morphological and anatomical characteristics**
- d) Genetic characters

3.4. References

1. http://www.brainkart.com/article/Types-of-classification---Taxonomy_32972/
2. <https://www.youtube.com/watch?v=KV0kVNP1OpY>
3. <https://www.preservearticles.com/botany/essay-on-the-natural-system-of-classification-of-plants/8696>

LESSON 4: DEVELOPMENT OF CLASSIFICATION SYSTEMS - PHYLOGENETIC

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4.1. Introduction

In this lesson we cover the phylogenetic classifications which were based on evolutionary and genetic relationship. These systems came after the publication of Charles Darwin's "Origin of Species" in 1859. Darwin believed in evolutionary processes ie that the present forms of life are products of evolution processes. The classifications that followed his theories therefore tried to establish ancestral relationships of plants.

4.2 Lesson Learning outcomes:

By the end of this lesson the learner will be able to:

4.2. 1. Describe phylogenetic classification systems

4.2.2 Distinguish primitive from advanced characters

4.2. 1. Describe phylogenetic classification systems

Augustin Wilhelm Eichler (1839-1887).

He was a German botanist and was the first person to construct a phylogenetic system. He divided the plant kingdom into two groups, Cryptogamae and Phanerogamae. Cryptogamae these are non seed plants which he divided into Thallophyta (algae and fungi), Bryophyta (Hornworts, Liverworts and Mosses) and Pteridophytes (ferns and fern allies). Phanerogams, the seed plants were divided into Gymnosperms and Angiosperms. Angiosperms were then divided into monocotyledons and dicotyledons. The dicotyledons were then divided into two groups, Chloripetalae (free petals) and sympetalae (fused petals). Eichler regarded the gymnosperms as more primitive than angiosperms and monocotyledons more primitive than dicotyledons.

Adolf Engler (1844-1930).

He was a German professor of botany at the University of Berlin and a Director of Botanic Gardens at Breslau. His first publication was a guide to the botanical gardens. He joined another botanist Eugen Prantl and together published a 20 volume work called *Die Natürlichen Pflanzenfamilien*. In this work all genera of the world from algae to seed plants were covered including an identification key to the genera. It was therefore widely accepted as it offered a means of identification for many plants. Many herbaria are arranged according to Engler and Prantl system e.g. Leningrad, Berlin, and Arnold arboretum. Various modern Floras are also arranged by the system including Flora Europaea.

Charles Bessey,

He was the first American to make a major contribution to plant classification. His classification was titled "The Phylogenetic Classification of Flowering Plants" in the *Annals of the Missouri Botanic Gardens*. His system was based on the Bentham and Hooker system. Which he rearranged to reflect evolutionary principles. Many concepts used today on primitive flowers are those of Bessey's e.g. primitive flowers have numerous perianth parts, many pistils, many stamens, superior ovaries and are insect pollinated. The flowers of the

family Magnoliaceae are still regarded as representative of primitive flowers. Most modern schemes are John improvement of Bessey's system.

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Hutchinson (1884-1972)

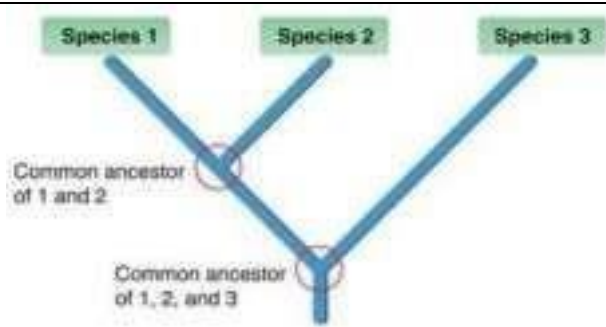
He was a British botanist and a former director of the Royal Botanic Gardens, Kew, England. His Phylogenetic system is published in "*The Families Flowering Plants*" (1959). He took up many of Bessey's ideas but differed especially in his division of the dicotyledons into woody and Herbaceous groups. He regarded the angiosperms to be a monophyletic group which arose from an ancestral group of Gymnosperms, a similar view to that of Eichler. The angiosperms had two lines of development hence the two divisions.

The woody group he called Division Lignosae and the herbaceous group the Herbaceae. According to him the Lignosae is more primitive than the Herbaceae meaning that it appeared earlier than the Herbaceae. Hutchinson has been criticized for the division because some families, which are very closely related on the basis of floral characters, are widely separated. For example, the herbaceous Lamiaceae from the woody Verbenaceae. Hutchinson attributed those characters due to convergent evolution that is similar as a response to selection pressures but not due to ancestral relationships. According to him the most primitive family among the Lignosae is the family Magnoliaceae and the most advanced Verbenaceae. Among the Herbaceae the most primitive family is Ranunculaceae and the most advanced Lamiaceae. Hutchinson considered the monocotyledon to have arisen from the dicotyledons, which means the monocots are more advanced than the dicotyledons (a view contrary to that of Eichler). His classification of the monocots is regarded as more natural than that of the earlier schemes.

E-tivities 4.2.1 Phylogenetic systems

Numbering, pacing and sequencing	4.2.2
Title	
Purpose	To describe the contributions to plant taxonomy by botanists who advanced phylogenetic systems of classifications
Brief summary of overall task	Watch video phylogeneic systems

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Spark	
Individual task	Outline the contribution to plant taxonomy by botanists who developed phylogenetic classification systems after Darwin's theories of evolution. Post the answer in the discussion forum
Interaction begins	Comment on the answers by your colleagues in the discussion forum
E-moderator interventions	1. Ensure that learners are focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress.
Schedule and time	This activity should take 1 hour

Next	Primitive and advanced characters
------	-----------------------------------

4.2.2. Primitive versus Advanced characters

Primitive characters are also referred to as unspecialized while advanced are also referred to as derived or specialized characters.

Primitive characters are said to be those of the common ancestor while the advanced ones are those that have departed from the ancestral conditions. Not all organs evolve at the same rate so that a family can have advanced and primitive characters. The lower Angiosperm families like the Magnoliaceae and Ranunculaceae have many primitive characters compared to the higher families like the Labiatae and Compositae. In the latter families, the ancestral features appear to have been lost.

Some of the morphological trends mentioned by Hutchinson regarding primitive and advanced characters in angiosperms include the following:

1. Trees and shrubs are probably more primitive than herbs.
2. Perennials are more primitive than biennials and annuals have been derived from both.
3. Aquatic plants are more advanced than related terrestrial plants

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4. Plants with vascular bundles arranged in a ring are more primitive than those with scattered vascular bundles.
5. Alternate leaves are more primitive than opposite and whorled types.
6. Compound leaves are more advanced than simple leaves.
7. Bisexual flowers are more primitive than unisexual ones and dioecious more advanced than monoecious ones.
8. Solitary flowers are more primitive than the inflorescence, the highest forms being the umbel and the capitulum.
9. Imbricate aestivation is more primitive than valvate one.
10. Many floral parts are more primitive than fewer parts.
11. Flowers with petals are more primitive than apetalous (without petals, like in the family Poaceae) ones.
12. Free petals (polypetalous) are more primitive than fused petals (syncarpous)
13. Regular flowers with radial symmetry (actinomorphic) are more primitive than bilaterally symmetrical (zygomorphic) flowers
14. Flowers with superior ovaries are more primitive than those with perigynous and epigynous ones
15. Free carpels are more primitive than syncarpous ones.
16. Numerous carpels are more primitive than fewer carpels.
17. Many stamens are more primitive than fewer stamens.
18. Fused stamens are more advanced than free ones.
19. Aggregate fruits (those formed from separate flowers) more advanced than single fruit
20. The capsule is more primitive than the drupe or berry

Additional trends not given by Hutchinson

- 21. Stipulate leaves are more primitive than exstipulate ones and free stipules more primitive than fused ones
- 22. Spininess of leaves and branches is an advanced character
- 23. Succulency is an advanced character.
- 24. Fibrous roots are more advanced than taproot

Some of the most advanced families are Rubiaceae and Compositae have many advanced characters and few primitive ones.

E-tivities 4.2.2 Primitive and Advanced characters

Numbering pacing and , sequencing	4.2.2
Title	
Purpose	To distinguish between primitive and advanced characters

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Brief summary	
of overall task	Watch video Primitive vs advanced characters

Spark	 Primitive Versus Advanced Floral Characteristics ■ Primitive ■ Flower parts many, more than ten ■ Flower parts free (not united or fused) ■ Superior ovary ■ Flowers with radial symmetry ■ Advanced ■ Flower parts few, ten or less ■ Flower parts united or fused ■ Inferior ovary ■ Flowers with bilateral symmetry
Individual task	Watch the video above and list the advanced and primitive characters. Post the answers in the discussion forum.
Interaction begins	Comment on the answers given by your colleagues in the discussion forum
E-moderator interventions	1. Ensure that learners are focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress.
Schedule and time	This activity should take 1 hour
Next	Taxonomic units of classification

4.3. Assessment Questions

1. The early phylogenetic systems were dominated by the:
 - a) Greeks
 - b) British
 - c) Americans
 - d) Germans

2. According to Hutchinson monocotyledons are more advanced than dicotyledons T / F
3. Adolf Engler and Eugen Prantl published:
 - a) *The Phylogenetic classification of Flowering Plants*
 - b) *Families of Flowering plants*
 - c) ***Die Naturlichen Pflanzenfamilien***
 - d) *Origin of Species*
4. Which of the following features is not an advanced character
 - a) Compound Leaves
 - b) Superior ovary
 - c) Opposite leaves
 - d) Few flower parts

4.4. References

1. Phylogenetic systems <https://www.youtube.com/watch?v=I5Wc31DQnhE>
2. Primitive vs advanced characters <https://www.youtube.com/watch?v=yn-9arDtNTY>

5.1 Introduction

In this lesson I will take you through the units of botanical classification. You have at some points in your study of botany, heard about the plant kingdom, the divisions, class, order etc.

These are some of the units of classification. In this lesson you will learn more about these units plus others you may not have come across.

5.2 Lesson learning outcomes;

At the end of the lesson you will be able to:

5.2.1 Arrange the units of classification in a hierarchical order

5.2.2 Describe the derivation of specific epithets

5.2.1 The taxonomic hierarchy.

Classification is about the placing of plants into groups according to their relationships. The plant groups are arranged in different levels from the lowest to the highest according to their magnitude. Therefore the groups are arranged in a hierarchical manner. The different levels are called categories and they constitute the units of classification. A tree diagram called a dendrogram usually illustrates the hierarchy, starting from the lower levels to the highest level which is the Plant Kingdom.

Species are combined into an ascending series of successfully larger and bigger categories, ultimately arriving at a single group covering all plants, the kingdom. Each taxon at the lower level can belong to only one taxon at the next higher level, for example a species can belong to only one particular genus. The units of classification are prescribed by the International Code of Nomenclature for Algae, Fungi and Plants. The code contains a set of rules of nomenclature to guide in the selection and application of names. The code prescribes the sequence and order of the units. The levels and the plant group assigned to them are named in the same way, thus a plant group at the genus level is a genus, a species level is a species so on.

The taxonomic hierarchy below is arranged in a descending order from the highest to the lowest:

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KINGDOM

DIVISION

CLASS

ORDER

FAMILY

Tribe

GENUS

Section

Series

SPECIES

Variety

Form

These are 12 main units prescribed by the code and 7 principal units as capitalized. The principle units cannot be omitted in any classification, even for the smallest group. For example the family Malvaceae must have genera and species. Apart from the principal units the others may be omitted if not required. The code also allows subcategories below each rank using the prefix sub added to the main category e.g. subdivision, suborder, subtribe, subsection etc. the additional units are provided for the sake of large groups like Compositae and Poaceae which show a lot of variation and which should be accommodated. Thus in these families there are subfamilies, tribes, genera, sections and so on.

The units of classification are arranged above in descending order from the highest to the lowest. The division is prescribed by the code to represent the category of highest magnitude within the plant kingdom (kingdom Plantae). The number of divisions within the plant kingdom varies with different systems of classification.

Names of taxa at different ranks

The names of taxa at the upper six **principle** ranks are single words or uninomials e.g. genera: *Combretum*, *Hibiscus* and *malva*. The other supraspecific ranks (other ranks above the species level but excluding the principle ranks) names are a combination consisting of the name of the taxon of the principle rank followed by the epithet of the lower rank They are joined by a word indicating the rank e.g. *Ranunculus* subgenus *Batrachium*.

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The code prescribes the endings to the latin names of some taxa to indicate their ranks.

Plant group	Ending	Example
Kingdom		Plantae

Subkingdom	-bionta	Embryobionta
Division	-phyta	Bryophyta
Class	-opsida	Angiospermopsida
Order	-ales	Malvales
Suborder	-ineae	Malvaceae
Family	-aceae	Malvaceae, Rosaceae
Subfamily	-oideae	Rosoideae
Tribe	-eae	Astereae
Subtribe	-inae	Potentillinae

. The Latin name of all the families end with - aceae e.g. Malvaceae and Euphorbiaceae. However, there are eight family names that do not end with aceae due to their long usage. These are very natural families whose names have been in existence for a long time and they are given exception to the rule but they have also been provided with alternative names ending with - aceae. These are:

Gramineae - Poaceae,

Cruciferae – Brassicaceae,

Leguminosae – Fabaceae,

Guttiferae – Clusiaceae,

Umbeliferae – Apiaceae,

Labiatae – Lamiaceae

Compositae – Asteraceae.

Palmae – Arecaceae

The family represents a more natural unit than any of the higher categories. It is difficult to recognize an order, this is because a lot is known about the components of a family and there are correlations of a large number of characters e.g. families such as the Gramineae, the Compositae and Labiatae are readily recognized as natural taxa whose respective members have definite characters binding them together.

Genus names

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The name of a genus is a single word, a noun in the singular number. It is the first name of a binomial e.g. *Hibiscus aponeurus*, the former name is the genus name. The initial letter of a genus name is written in capital and always italicized

Names of species.

The name of a species is a binary combination consisting of the name of the genus and the specific epithet.

If an epithet has two or more names they are united or hyphenated so that they appear as one e.g. *Hibiscus rosasinensis* or *Hibiscus rosa-sinensis*. The specific epithet cannot repeat the genus name and the species name is italicized

Infraspecific taxa

Names below the species level are known as infraspecific taxa. The names of infraspecific taxa are trinomials that consist of three names, the name of the species and an infraspecific epithet connected by a term showing its rank e.g. *Salvia grandiflora* subsp. *williana*. Note the infraspecific name is italicized. The infraspecific epithet can sometimes repeat the specific epithet, e.g. *Lobelia spicata* var. *spicata*

Note: All specific and infraspecific epithets must be written in small letters

E-tivity 5.2.1 The taxonomic hierarchy

Numbering, pacing and sequencing	5.2.1
Title	
Purpose	To familiarize with the rules of nomenclature on taxonomic units of classification
Brief summary of overall task	Read: Taxonomic hierarchy Watch video: Taxonomic units
Spark	

	
Individual task	Explain the taxonomic hierarchy and the endings used on names of the units of subdivision and subclass giving examples. Post your answers in the discussion forum.
Interaction begins	Comment on the answers given by your colleagues in the discussion forum
E-moderator interventions	1. Ensure that learners re focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress. 4. Close the discussions
Schedule and time	This activity should take 1 hour
Next	Source of specific epithets

5.2.2. Derivation of specific epithet

Specific epithets can be derived from three sources:

Commemorative epithets:

These are used to honour or commemorate someone e.g. the one who discovered the new species, or one who has done a lot of taxonomic studies in the group or collected a lot of specimen of that species. *Hibiscus greenwayi* in honour of Greenway, *Aloe petersonii* in honour of Peter.

Geographical epithets

These are derived from geographical locations
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Examples: Hibiscus rosa-sinensis from China. Vitex keniensis from Kenya

Senecio ruwenzoriences from the Ruwenzori Mountains, Aloe rhodesiana from Rhodesia

Descriptive epithets

These describe characteristics of the plant species e.g. Gossypium hirsutum (hairy character), Mentha aquatic, grows in aquatic habitats, Croton megalocarpus (Croton with large fruits)

Specific epithets have Latin endings. Every Latin or Latinized nouns have gender, masculine (m) or feminine (f) or neuter (n). Each gender has an ending:

a -feminine

us-

masculine

um – neuter.

Specific epithet should agree in gender with the generic name

e.g. Lathyrus hirsutus, Gossypium hirsutum, Euphorbia hirta

E-tivity 5.2.2 Sources of specific epithets

Numbering, pacing and sequencing	5.2.2
Title	
Purpose	To describe sources of specific epithets
Brief summary of overall task	Watch video: genus and species names

Spark	Meanings of Specific Epithets <table> <tr> <td><i>arvensis</i></td><td>growing in a field</td></tr> <tr> <td><i>esculentus</i></td><td>edible</td></tr> <tr> <td><i>officinalis</i></td><td>used medicinally</td></tr> <tr> <td><i>palustris</i></td><td>growing in a marshy place</td></tr> <tr> <td><i>arenicola</i></td><td>growing in sandy soil</td></tr> <tr> <td><i>salina</i></td><td>growing in a saline environment</td></tr> <tr> <td><i>australis</i></td><td>southern</td></tr> <tr> <td><i>borealis</i></td><td>northern</td></tr> <tr> <td><i>orientalis</i></td><td>eastern</td></tr> <tr> <td><i>occidentalis</i></td><td>western</td></tr> </table>	<i>arvensis</i>	growing in a field	<i>esculentus</i>	edible	<i>officinalis</i>	used medicinally	<i>palustris</i>	growing in a marshy place	<i>arenicola</i>	growing in sandy soil	<i>salina</i>	growing in a saline environment	<i>australis</i>	southern	<i>borealis</i>	northern	<i>orientalis</i>	eastern	<i>occidentalis</i>	western
<i>arvensis</i>	growing in a field																				
<i>esculentus</i>	edible																				
<i>officinalis</i>	used medicinally																				
<i>palustris</i>	growing in a marshy place																				
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<i>salina</i>	growing in a saline environment																				
<i>australis</i>	southern																				
<i>borealis</i>	northern																				
<i>orientalis</i>	eastern																				
<i>occidentalis</i>	western																				
Individual task	Write 10 species names and determine the source of the specific epithet. Do this in the discussion forum																				
Interaction begins	Comment on the answers given by your colleagues in the discussion forum																				

E-moderator interventions	1. Ensure that learners are focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress. 4. Close the discussions
Schedule and time	This task should take 1 hour
Next	Characters and character states

5.3. Assessment Questions

1. Names of taxa above species level (supraspecific) are:

- a) Uninomials
- b) Binomial
- c) Trinomial
- d) Polynomial

2. The other name for the family Compositae is Liliaceae T/ F

3. In the case of *Aloe elgonica*, the specific epithet is derived from:

- a) A characteristic of the plant.
- b) Commemoration of someone
- c) A geographical location
- d) From the ecological habitat.

4. Names of genera have no specific ending, T / F

5.4. References

1. http://www.brainkart.com/article/Taxonomic-Hierarchy_32964/

2. Taxonomic units:

<https://www.youtube.com/watch?v=8QjBlNv76Y0>

3. Genus and species names:

<https://www.youtube.com/watch?v=EQnVIZa05ck>

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LESSON 6 CHARACTERS AND CHARACTER STATES

6.1 Introduction

In this lesson I will introduce you to the topic of characters and character states, both attributes used by plant taxonomists in carrying out taxonomic works. Characters have many definitions; I will define a taxonomic character as any feature of a plant that is divisible in at least two conditions or states. The following are examples:

Petal colour is a character, the states being white, yellow, orange and blue. Leaf arrangement is another character; the states are opposite, alternate and whorled.

Leave shape is a character; the states are linear, lanceolate, ovate, obtuse etc.

6.2 Lesson learning outcomes:

At the end of this lesson, you will be able to:

6.2.1 Discuss characters, character states, uses and types.

6.2.2 Explain sources of taxonomic characters

6.2.1 Characters and character states, uses and types

Uses of characters and character states:

Identification: For this the taxonomist uses distinguishing characters, that is those characters of limited occurrence selected so that they can be used to differentiate between different taxa. They are called diagnostic or key characters because they are used for constructing identification keys in taxonomic works in Floras, monographs, revisions and other taxonomic works..

Descriptions in Floras and other taxonomic books and works on the characteristics of plant groups.

Construction of classification systems for example the Hutchinson phylogenetic system, Bentham and Hooker's natural system.

Interpretation of evolutionary relationships. For Phylogenetic studies characters of the fossils materials are studied.

Qualitative and quantitative characters

Quantitative characters are those that can be assessed by counting or measuring and the stated given in terms of numbers; examples are:

- Leaf length with states given in cm.
- Number of stamens with states in numbers e.g. 2, 4, 6.
- Height of the plant with the states in numbers e.g. 1m, 1.5m. 2m

Qualitative characters describe form and shape and the states are given in descriptive terms; examples are:

- Ovary position, the states being superior and inferior.
- Habit of the plants; states are trees, shrubs, and herbs.
- Leaf margin; states are entire, serrate, crenate, and lobed.

Good and bad characters

The terms good and bad are applied to characters of lesser or greater value. Similar pairs of terms are reliable and unreliable characters or meaningful and meaningless characters. The terms are relative because what may be useful characters in one group may not be useful in another group. In general, good characters have the following attributes:

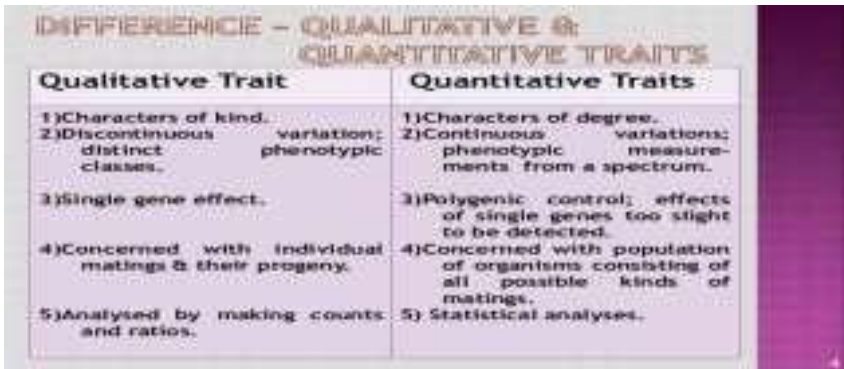
- 1. Are not subject to wide variation within the samples being considered.
- 2. Do not have high genetic variability.
- 3. Are not susceptible to environmental modifications.
- 4. Show consistency with the correlation of other characters previously used in delimiting a natural group (agree with a natural classification constructed without their use).

Bad characters lack the above qualities

E-tivities 6.2.1 Characters and character states

Numbering, pacing and sequencing	6.2.1
Title	
Purpose	To distinguish between characters and character states and their uses
Brief summary of overall task	types of characters

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Spark	
Individual task	List four qualitative characters and their states List four different quantitative characters and their states Answers to be written in the discussion forum
Interaction begins	Comment on the answers given by your colleagues in the discussion forum
E-moderator interventions	1. Ensure that learners are focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress. 4. Close the discussions
Schedule and time1.	This task should take 1 hour
Next	Sources of taxonomic characters

6.2.2 Sources of taxonomic characters

Morphology

Morphological features are the most used for constructing classification systems because they are easily observed with unaided eyes, with the use of a hand lens or with a binocular microscope. They are readily appreciated, for example one can easily appreciate a red flower rather than the number of chromosomes which you obtain with difficulty. Morphological characters include floral, fruit and vegetative characters.

a) Floral characters include characters like flower shape, petal number, aestivation, cohesion of stamens, ovary position etc. Floral characters tend to vary little between closely related

taxa and are not influenced by environmental conditions. They are therefore reliable and relatively conservative and are the most used in the classification of angiosperms.

b) Fruit and seed characters are less reliable than floral characters but are very useful in some groups, for example Brassicaceae and Apiaceae. In the Caryophyllaceae seed characters are useful at all levels of the taxonomic hierarchy.

c) Vegetative characters include characters of the leaf and stem. Leaf characters include, presence of stipules, leaf arrangement, leaf shape (linear, lanceolate, ovate, cordate, spatulate) leaf apices (acute, acuminate), leaf venation etc. Stem characters include stem shape (round, square) e.g. the Labiatae family is characterized by square stems. Stem habit (tree, shrub or herbs), root characters are useful but employed in broad terms such as fibrous and taproot.

Compared to floral characters, vegetative characters are less reliable because they are easily influenced by environmental factors but in some genera vegetative characters of the leaf provide the most useful characters. Vegetative characters are especially useful in the identification of non-flowering material such as fossils.

Anatomy

Anatomical data such as wood anatomy is used to solve taxonomically difficult groups. Other data like trichomes (hairs) and stomata characters are used for routine identification. For trichomes the different types are studied. Leaf epidermis both abaxial and adaxial sides. Anatomical features are useful at all levels of the taxonomic hierarchy. Leaf epidermis yields many useful characters such as the distribution of the stomata and arrangement of the subsidiary cells surrounding the stomata. In the family Poaceae (grasses) the flowers are much reduced and provide a limited number of taxonomic characters; anatomical characters of the leaf epidermis and leaf anatomy provide extensive taxonomic data. Characters such as the disposition of the sclerenchyma, the arrangement and form of the vascular bundles, the differentiation of epidermal cells and form and distribution of silica bodies and various types of trichomes and papillae have played a big part in the re-classification of the family from the level of subfamily to species.

Palynology

This is the study of pollen grains. Pollen grains are well preserved in the herbarium specimens and also in fossils. Pollen preserved over the years under lakes is used to study past vegetation. Pollen offers many characters of taxonomic value. Pollen grains are very small and need to be studied by using microscopes. The compound microscope can be used to assess some features but electron microscopes (SEM and TEM) with their high magnifications and high resolution power reveal many characters of taxonomic value

Pollen data is useful at all levels of the taxonomic hierarchy and include:

- a) Size and aggregation of the grains: into tetrads as in Ericaceae, Polyads as in *Mimosa* and Orchidaceae (known as pollinia in Orchids, all the pollen grain of each lobe of the anther aggregate together) etc.

- b) Apertures: Some pollen grains have pores (porate) while others have furrows (corporate). Aperture numbers may vary in different plant groups for example some grains are monocolpate, monoporate, tricolpate, tetracolpate, hexacolpate etc.

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- c) Surface sculpturing (ornamentation), for example Poaceae pollen grains is smooth while in the Asteraceae the pollen grains are spiny
- d) Internal wall structure of the exine (pollen grains have two walls, the intine is the innermost and the exine is the outermost). It is only the internal structures of the exine that offer characters of taxonomic value

Pollen grain characters may be more useful in some plant groups than in others, for example in the Poaceae (Gramineae) pollen grains are very uniform and offer few valuable characters while in the family Acanthaceae they provide many useful characters at all levels from subfamily and below.

Chemical

Chemical characters have been used in classifications. The data includes presence and absence of chemical compounds, extraction, separation and identification. Some of the compounds used are proteins, nucleic acids, polysaccharides, lipids, glucosides, essential oils detected as odour in some species of Lamiaceae and Apiaceae. The most used compounds in taxonomy are the flavonoids. These are low molecular weight secondary plant products. These compounds are not affected by preparation of herbarium specimens hence their great use. There are about 500 different flavonoid compounds found in plants offering many variable characters.

CYTOLOGY

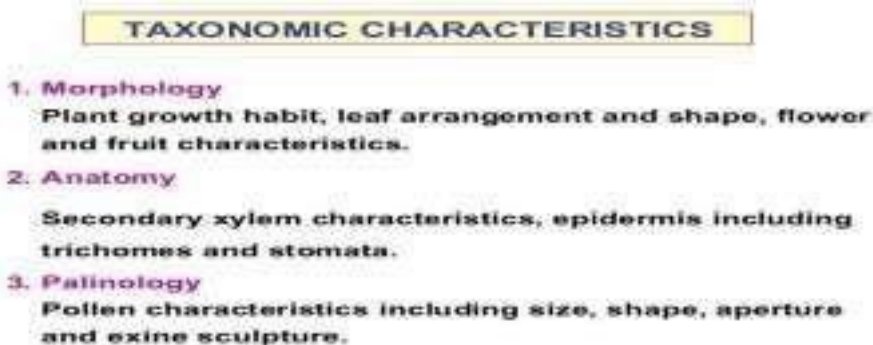
Cytological characters are mostly from the nucleus. These are mainly chromosome morphology, size and shape, chromosome number and pairing at meiosis. The most recent studies include gene sequencing of the DNA molecule especially the chloroplast DNA for phylogenetic studies.

E-tivity 6.2.2 Sources of taxonomic characters

Numbering, pacing and sequencing	6.2.2
Title	
Purpose	To discuss the different sources of taxonomic characters

Brief summary of overall task	Read: morphological characters Watch the videos: anatomical characters Palynology chemotaxonomy Cytology
-------------------------------	--

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Spark	
Individual task	After reading and watching the videos, outline the sources of taxonomic characters. Do this in the discussion forum
Interaction begins	Comment on the answers given by your colleagues in the discussion forum
E-moderator interventions	1. Summarize the threads 2. Give feedback 3. Provide teaching points 4. Close the E-tivity
Schedule and time	This task will take 2 hours
Next	Nomenclature

6.3. Assessment Questions

1. Identification characters are any attributes of a plant T / F
2. The number of stamens in a flower is a qualitative character T / F
3. Wood anatomy is used in problematic taxonomic groups T / F
4. Gene sequencing of the DNA molecule has proven valuable in interpreting evolutionary relationships T / F

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6.4. References

1. Types of characters https://www.youtube.com/watch?v=dgCHWg3fg_Y
2. Morphological characters <https://www.biologydiscussion.com/plant-taxonomy/morphology-and-plant-taxonomy/30414>
3. Anatomical: <https://www.youtube.com/watch?v=AO4Q3MQghfk>
4. Palynological: <https://www.youtube.com/watch?v=LR4TUZgMs6g> 5.
- Chemotaxonomy: <https://www.youtube.com/watch?v=sElH6Y7ipHM> 6.
- Cyology: https://www.youtube.com/watch?v=rA1stFMP3_o

LESSON 7 NOMENCLATURE**7.1 Introduction**

The naming of groups of plants and the rules governing the application of the names is called nomenclature. After classifying plants into groups, they are given names to facilitate communication. Plants have been given names since prehistoric times in local or vernacular names and these names are not understood by international community and have their disadvantages.

Disadvantages of common or vernacular name:

1. They may be quite indefinite that is many names are applied to the same species, for example Black jack, *Bidens pilosa* has many vernacular names depending on the community one comes from. Even within the same community, same name may be applied to very different species depending on the locality one comes from.
2. The use of the names is restricted to people who speak the same language.
3. The names are not regulated by any rules.

Advantages of scientific names:

1. They are governed by the nomenclatural rules e.g. The International Code of Nomenclature for Algae Fungi and Plants for wild plants.
2. They indicate generic relationships that is they are of biological significance. Related species are grouped under one genus, e.g. *Hibiscus rosa-sinensis*, *Hibiscus diversifolius*, *Hibiscus greenwayi*, *Hibiscus aponeurus* etc. are species of the same genus, *Hibiscus*
3. They are of international scope
4. Botanical names are based on the Latin language, which employs the Roman alphabet. This avoids the confusion that would result if the names were written indiscriminately in the characters of other alphabets e.g. Greek, Chinese, etc.

7.2 Lesson learning outcomes:

At the end of the lesson the learner will be able to:

7.2.1 Describe the history of nomenclature and the rules of nomenclature on Typification

7.2.2 Describe the conditions for valid publications, synonyms, citation of author names and conditions for name changes

7.2.1 History of nomenclature and Typification

Historical background of organized nomenclature

Before the middle of 18th century the names of plants were a mixture of monomials, binomials, trinomials and more frequently cumbersome polynomials composed of several words. Modern botanical nomenclature starts with the publication of Linnaeus's *Species*

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Plantarum (1753). In this work Linnaeus started the consistent use of the binomial system to name plants.

The first rules on nomenclature were written by A.P. De Candolle in *Theories Elementaire De la Botanique* in 1813. The first organized efforts internationally to produce a nomenclatural system was made at the First International Botanical Congress held in 1867 in Paris called the Paris Code adopted from Alphonse De Candolle. From then on there has been many International Botanical Congresses (IBC) and the present rules are called The International Code of Botanical Nomenclature for Algae, Fungi and Plants published in 2012, passed at the 18th IBC in Melbourne, Australia. The code contains Rules and Recommendations on all points connected with naming plants. The present Code is the Shenzhen code passed in July 2017 at the 19th IBC in China and published in June 2018

Typification

The correct application of names of taxa of the rank of family and below is determined by nomenclatural types. The purpose of typification is to permanently fix the application of names to prevent the possibility of the same name being used for different plants.

A nomenclatural type is that element of a taxon to which the name of the taxon is permanently fixed

Kinds of Types

The code recognizes the following nomenclatural types

Holotype: It is the one specimen or element designated by the author as the nomenclatural type.

Isotype: It is a duplicate of the holotype e.g. if several similar herbaceous plants are mounted on separate sheets or several branches of a tree are collected at the same time, one is designated as the holotype and the others become isotypes. The isotypes are sent to different herbaria.

Syntypes: These are specimens cited by the author when no holotype was designated at the time of publication.

Lectotype: It is a specimen selected from the original specimens (syntypes) to serve as a nomenclatural type when no holotype was designated at the time of publication or as long as it is missing or lost.

Neotype: It is a specimen selected to serve as the nomenclatural type as long as all of the material on which the description of the new species was based is missing or lost. This is not really a type but is a specimen chosen to serve as the standard reference point. Such a specimen should match the original description and if possible come from the same locality as the holotype.

Effects of types on species names:

When the rank of a taxon is changed, or it is transferred it automatically retains the same type

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The name of an infraspecific taxon which retains the type specimen of the species must have the same epithet as the species but no author citation e.g. *Glycine wightii* (Wight & Arn.) Verdc. Has two subspecies

G. wightii (Wight & Arn.) Verdc. subsp. *wightii*.

G. wightii (Wight & Arn.) Verdc. Subsp. *Pseudojavonica* (Taub.) Verdc.

The former subspecies retains the type specimen of the species and is known as typical or nominate species. Repeated names, the specific epithet and the infraspecific epithet above are called autonyms

Description of a new species quoting the types:

When a new taxon is described, the name is usually indicated by sp. nov.

(species nova). Example: *Gaudinia hispanica* Stace and Tutin, (sp. nov).

Description is in Latin as below:

Gramen annuum, 6-28 cm altum, inflorescentia 1-14 cm. longa, erecta, rigid
.....etc.

Holotypus: Near Ermito del Rocio, S.W of Servilla, Huelva, Spain. 25 May 1967 Moore and Tutin, S.N, 1280. (K) Isotypi (K). S. Silvestre and B. Valdes 2289/ 69. (K is for Kew herbarium)

E-tivity 7.2.1 Typification

Numbering, pacing and sequencing	7.2.1
Title	
Purpose	To describe the rules of nomenclature regarding typification

Brief summary of overall task	Read: vernacular vs scientific names typification
-------------------------------	---

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Spark	
Individual task	Read and outline the different kinds of types and post in the discussion forum
Interaction begins	Comment on the answers given by your colleagues in the discussion forum
E-moderator interventions	1. Summarize the threads 2. Give feedback 3. Provide teaching points 4. Close the E-tivity
Schedule and time	This task will take 1 hour
Next	Other rules of nomenclature, Svalid publications, synonyms, citation of authors names

.7.2.2 Conditions for valid publications, synonyms, citation of authors names and causes for name changes.

Conditions for valid publications

1. Be effectively published – printed in a book or journal or electronic publication. 2. Be accompanied by a diagnosis in Latin or English or by reference to a previously and effectively published diagnosis.
3. Be accompanied by a nomenclatural type if it is of the rank of family and below.

SYNONYMS

These are different names for the same species. In practice the term is applied only to names other than the considered correct name:

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Example: *Eulophia streptopetala* Lindley

Synonyms: *E. grantii* (Reichb.f.) Summerh.

E. paivaeana (Reichb. f.) Summerh.

There are two types of synonyms:

Nomenclatural or homotypic synonyms – based on the same type, for example due to transfer of a species from one genus to another e.g. *Sisymbrium murale* transferred to *Diploaxis muralis* (L). DC

Taxonomic or heterotypic synonyms. These are based on different types from the correct name so that their identity is a matter of taxonomic judgment e.g. *Ruppia maritima* L. (1753) and *Ruppia rostellata* Koch. (1824) are synonyms based on different types.

Citation of author's names

For a plants name to be accurate and complete, it should be followed by the name of the publishing author. This enables one to trace the original description and ascertain its nomenclatural type and date of publication. Sometimes complete citation is given showing the place of publication e.g. *Ranunculus acris* L., *Species plantarum* pg 554, 1754. If the publication was not entirely written by the author of the name the word in is used for example *Ramatouella virens* Spruce ex Eichler in Martius. *Flora Brasiliensis* (14(2): 100(1867)).

Botanical names are published in *Index Kewensis*. Sometimes an author takes up an unpublished name for a new species and validly publishes it with his own description. In that case 'ex' is usually inserted between the man who suggested the new name and his own as the publisher e.g. *Capparis lasianthe* R. Brown ex DC.

This means that Robert Brown who gave the new name but did not publish it, but it was De Candolle who validly published the new species using Brown's name.

CAUSES FOR NAME CHANGES:

Principle of Priority

Each taxon has only one correct name, the earliest that is in accordance with the rules. For example Bentham and Hooker recognized two species *Ruppia rostellata* Koch (1824) and *Ruppia maritima* L. (1753). Later examination of the nomenclatural types of these two species established that Koch's *Ruppia rostellata* was identical to Linnaeus's *Ruppia maritima*. According to the principle of priority, *R. maritima* which is the earlier name is the correct name.

Transference:

A species may have been described in a particular genus and transferred to a different genus. When this happens the name of the original author is put in brackets, followed by the name of the author who effected the transfer e.g. *Sisymbrium murale* L. was transferred by De Candolle to *Diploaxis*. It is therefore written as *Diploaxis muralis* (L.) DC. By means of the double citation the original description, hence the type specimen can be traced. A variety may be raised to specific rank e.g. *Crocus sativus* L var. *vernus* L becomes *Crocus vernus* (L.) Hill. Again this shows a change on the species in which Linnaeus is the original author and Hill having effected the change.

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When a species is transferred to another genus without changes of rank, the original specific epithet is to be retained e.g. *Sisymbrium murale* L. which when transferred to *Diploaxis* retains its specific epithet and becomes *Diploaxis muralis* (L.) DC. This is so unless in the new position, there is another species bearing the same name.. In such a case a new name must be found for the transferred species to prevent homonymy (same name for different species) which can lead to confusion. Later homonyms are illegitimate e.g. *Spartium biflorum* Desf. (1798) was transferred by Spach to *Cytissus* in 1848, but as there was already a species called *Cytissus biflorus* L'Heritier (1791) Spach had to give a new name to the transferred species *Cytissus fontanessii* to Desfontaines species.

REJECTION OF NAMES

When two or more species are united into one, the oldest legitimate name is retained for the united species according to the principle of priority. If the names of the species are of the same date, the author who unites them has the right to choose any of the names. For example there were two species described by Linnaeus, *Waltheria indica* L. (1753) and *Waltheria americana* L. (1753) at the same time in the Species Plantarum. Robert Brown united the two species and chose *Waltheria indica* for the combined species. The other name *Waltheria Americana* is rejected.

A name cannot be rejected because it is inappropriate e.g. *Scilla peruvian* is the name of a Mediterranean species but it is not rejected just because the name implies the species name is derived from Peru which is a country from South America.

A name must be rejected if it is a later homonym. The South African *Impatiens capensis* Thunberg (1794) is a later homonym of *Impatiens capensis* Meerburg (1775) i.e. there are different species with the same name. *I. capensis* Thunberg must therefore be rejected and another name found for the species which was *I. dulthiae* L. Borus, 1921)

NAME CHANGES

It can be concluded that many name changes are due to application of the rules.

Application of the principle of priority (if the name used is not legitimate)

Due to changes in taxonomic opinion – for example, amalgamation, splitting or transference of taxa

Changes due to homonymy

Other matters treated by the code.

When describing a new species a name must be found which has not been used in the genus before.

Orthography. Plant names must be spelt as they were originally published unless the original spelling was definitely wrong or misprinted

Synonyms are best cited chronologically (in order of occurrence, the oldest first

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E-tivity 7.2.2 Conditions for valid publications, synonyms, and citations of authors names

Numbering, pacing and sequencing	7.2.2																		
Title																			
Purpose	To describe the nomenclatural rules regarding valid publications and citation of authors names																		
Brief summary of overall task	Watch video: Citation of author's names																		
Spark	<table border="1"> <thead> <tr> <th>Author</th><th>Standard form of Abbreviation</th></tr> </thead> <tbody> <tr> <td>Linnaeus</td><td>L.</td></tr> <tr> <td>G.Bentham</td><td>Benth.</td></tr> <tr> <td>William Hooker</td><td>Hook.</td></tr> <tr> <td>Robert Brown</td><td>R.Br.</td></tr> <tr> <td>J.P.Lamarck</td><td>Lamk.</td></tr> <tr> <td>A.P.de Candolle</td><td>DC.</td></tr> <tr> <td>Wallich</td><td>Wall.</td></tr> <tr> <td>Alphonse de Candolle</td><td>A. DC.</td></tr> </tbody> </table>	Author	Standard form of Abbreviation	Linnaeus	L.	G.Bentham	Benth.	William Hooker	Hook.	Robert Brown	R.Br.	J.P.Lamarck	Lamk.	A.P.de Candolle	DC.	Wallich	Wall.	Alphonse de Candolle	A. DC.
Author	Standard form of Abbreviation																		
Linnaeus	L.																		
G.Bentham	Benth.																		
William Hooker	Hook.																		
Robert Brown	R.Br.																		
J.P.Lamarck	Lamk.																		
A.P.de Candolle	DC.																		
Wallich	Wall.																		
Alphonse de Candolle	A. DC.																		
Individual task	Describe the citation of author's names and the current rules on valid publications. Do this in the discussion forum																		
Interaction begins	Comment on the answers given by your colleagues in the discussion forum																		

E-moderator interventions	1. Summarize the threads 2. Give feedback 3. Provide teaching points 4. Close the E-tivity
Schedule and time	This task will take 1 hour
Next	Identification

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7.3. Assessment Questions

1. Which of the following statements is not true about scientific names?
 - a) They are restricted to people who speak the same language
 - b) They are of biological significance
 - c) They are regulated by nomenclatural rules
 - d) They are based on the Latin Language
2. Typification applies from the rank of family and below T / F
3. A lectotype is a specimen designated by the author as the nomenclatural type at the time of publication T/ F
4. Synonyms are different names for the same species T / F

7.4 references

1. Vernacular vs. scientific names <https://www.slideshare.net/bisharifa/botanical-nomenclature>
2. <https://www.slideserve.com/ewa/typification-icbn-art-7-10>
3. Typification: <https://www.youtube.com/watch?v=5eNe82l5uso>
4. Citation of author's names <https://www.youtube.com/watch?v=0Rp4xEvsrxA>

LESSON 8: IDENTIFICATION AND CLASSIFICATION OF CULTIVATED PLANTS**8.1 Introduction**

Identification is the determination of a taxon as being identical with or similar to another and already known element. The determination may be arrived at by the aid of literature or by comparison with plants of known identity. In some instances the plant may be determined as new to science. It is customary in taxonomic practice to combine identification with the determination of the correct scientific name.

8.2. Lesson learning outcomes:

By the end of this lesson the learner will be able to:

8.2.1 Describe the methods for plant identification

8.2.2 Describe the use of identification Keys

8.2.3 Discuss the classification of cultivated Plants

8.2.1 Methods of plant Identification

Plant identification is accomplished by one or more methods or combination of methods. The first step with unknown species is to determine the family to which the plant belongs. For this you use the family key. Family keys are provided for in several books e.g. Hutchinson's "*Families of Flowering Plants*" or when the unknown plant is of a known locality, the regional flora e.g. *Flora of Tropical East Africa* for East African plants. Knowing the family, one turns to the section where it is treated and there by means of key to the genera repeats the procedure to determine the generic name.

To identify the Species

To identify the species several methods are used:

- Keys to Floras, monographs and revisions.
- Illustrations and Photographs
- Herbarium specimens

Keys

To identify a species one uses the key to the species of a particular genus in a Flora, monograph or revision. By use of the key one also knows the scientific name of the species. The identity of a plant obtained solely by use of the key may not be correct. It is important to confirm the identity by reading the description provided to ensure that the characters observed in the unknown plant matches the description. When a marked variance exists between the two a misidentification probably exists. A plant may not be determined by using the key if it is new. The more one uses keys the easier it becomes to understand the characters hence identification.

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Flora is an account of plants occurring in a particular area for example the *Flora of Tropical East Africa*. The plants or vegetation is referred to as flora (f in the lower case).

Monographs and Revisions A monograph is a comprehensive treatment of a group of plants. The treatment is a synthesis of all that is known about the group. It involves the incorporation of the previously known information with the monographers own. It's geographical scope is worldwide e.g. a monograph of *Nicotiana*, *Pinus* or *Datura* would include complete synonymy, detailed description, detailed listing of ecological, geographical, cytological, chemical, anatomical and any other data.

A taxonomic revision differs from a monograph in the degree of scope and completeness. Revision is confined to taxonomic and geographic information. It usually includes a complete synonymy, short descriptions often confined to diagnostic characters. Other types of data such as ecological, chemical, genetic etc. are not included. Revisions are done on regional or worldwide basis.

Illustrations and Photographs

These are usually referred to when specimens have not been seen but they can be used in conjunction with other methods

Characters Considered Before Plant Identification:

- i. Whether a plant is herbaceous or woody, and annual or perennial in nature.
- ii. Whether or not milky or coloured sap is present in the leaf, stem or other plant parts.

- iii. The leaf type, phyllotaxy, and venation.
- iv. Presence or absence and type of stipule on young shoots.
- v. The distribution and kinds of surface coverings (i.e. hairs, trichomes, spines, etc.).
- vi. The parts of the flower and the number of sepals and petals, whether separate or fused, and also their arrangement i.e. aestivation.
- ii. Whether perianth is present in one or more series, or absent.
- viii. Whether pappus (e.g. Asteraceae) or epicalyx (e.g. Malvaceae) or similar structures are present.
- ix. Whether a nectar-secreting disc is present in the flowers (e.g. Rutaceae).
- x. Whether the flowers are actinomorphic or zygomorphic.
- xi. The number and attachment of stamens and if there is any fusion of anthers or filaments.
- xii. The number of pistils, styles and stigmas of the gynoecium, observation of a transverse section of the ovary, the number of locules, number of ovules per locule, and also the placentation.
- xiii. The position of the ovary and fusion of the perianth by observing a longitudinally cut section of the entire flower through its centre.

Plant Identification Methods:

The methods of identification include the following:

(a) Expert Determination:

The best method of identification is expert determination in terms of reliability or accuracy. In general the experts have prepared treatments (monographs, revisions, synopses) of the group in question, and it is probable that the more recent floras or manuals include the expert's concepts of taxa.

Experts are typically found in botanical gardens, herbaria, museums, colleges, universities, etc. However, although of great reliability, this method presents problems of requiring the valuable time of experts and creating delays for identification.

(b) Recognition:

It approaches expert determination in reliability. This is based on extensive, past experience of the identifier with the plant group in question. In some groups this is virtually impossible.

(c) Comparison:

A third method is by comparison of an unknown with named specimens, photographs, illustrations or descriptions. Although this is a reliable method, it may be very time consuming or virtually impossible due to the lack of suitable materials for comparison. The reliability is, of course, dependent on the accuracy and authenticity of the specimens, illustrations, or descriptions used in the comparison.

(d) The Use of Keys and Similar Devices (Synopses, Outlines, etc.):**E-tivity 8.2.1 Methods of plant identification**

Numbering, pacing and sequencing	8.2.1
Title	
Purpose	To describe the various methods used in plant identification
Brief summary of overall task	Read.: identification methods
Spark	

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Individual task	Outline the methods used in plant identification. Post your answer in the discussion forum
Interaction begins	Comment on the answers given by your colleagues in the discussion form

E-moderator interventions	1. Summarize the threads 2. Give feedback 3. Provide teaching points 4. Close the E-tivity
Schedule and time	This task will take 1 hour
Next	Use of identification keys

8.2.2 The use of Identification Keys

A key is an artificial analytical device or arrangement used for identification of unknown plants. The most used key is a dichotomous key, where two contradictory statements known as leads are given. The two contrasting statements make up a couplet. The two statements i.e leads of the same couplet must contrast (contradict each other). The two leads should start with the same word e.g. flowers or leaves. Two consecutive couplets should not begin with the same word to avoid confusion. A key uses diagnostic characters these are the distinguishing characters in a plant or plant groups, also called key characters.

Characters can be drawn from any part of the plant, but the most constant and easily observed characters should be chosen in preference to variable and more hidden characters or those observed with difficulty. Qualitative characters should be chosen in preference to quantitative ones. Where measurements are used, they should not overlap; example

Flowers red, 4-6 cm long.

Flowers yellow 6-8 cm long

Note the overlap in the measurements.

Bracketed and Indented keys:

There are two types of keys, bracketed and indented keys. In a bracketed key the two leads of each couplet are always together. In an indented key all the possibilities that depend on one lead are dealt with before the second one is taken up. Indented keys have the advantage of showing the visual presentation of the group so that one can get a picture of the taxa. They have the disadvantage of taking up more space than bracketed keys. Both keys are constructed using the same data.

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Examples of part of keys of British species of *Primula*

(a) *Bracketed key*

1. Leaves mealy beneath; Flowers lilac or purple . . . 2 _ leaves not mealy beneath;

Flowers yellow3 2. (1) Leaves margins crenulate; flowers lilac; heterostylous;

corolla lobes distant; calyx teeth $\pm$ acute; capsule cylindrical

much exceeding calyx 1. *farinosa*

_ Leaves margins not crenulate; flowers blue-purple,

not heterostylous; corolla lobes contiguous;

calyx teeth obtuse; capsule ovoid, slightly

exceeding calyx 2. *scotica* (b) *Indented*

key

1. Leaves mealy beneath; flowers lilac or purple 2. Leaves

crenulate; flowers lilac, heterostylous;

corolla lobes distant; calyx teeth $\pm$ acute; capsule

cylindrical, much exceeding calyx . 1. *farinosa*

2. Leaves not crenulate; flowers blue-purple, not

heterostylous; corolla lobes contiguous; calyx

teeth obtuse; capsule ovoid, slightly exceeding calyx

..... 2. *scotica*

1. Leaves not mealy beneath; flowers yellow

A DICHOTOMOUS KEY TO SELECTED GENERA OF SAXIFRAGACEAE

- 1. Shrub or woody vine 2.
- 1. Herbs 6.
 - 2. Woody vine; petals 7 or more Decumaria.
 - 2. Shrub; petals 4 or 5 3.
- 3. Leaves alternate or on short spur branches 4.
- 3. Leaves opposite 5.
 - 4. Leaves pinnately veined; ovary superior; fruit a capsule Itca.
 - 4. Leaves palmately veined; ovary inferior; fruit a berry Ribes.
- 5. Petals usually 4; stamens 20-40; fruit longitudinally dehiscent, not ribbed Philadelphus ●
- 5. Petals usually 5; stamens 8-10; fruit poricidally dehiscent, 10-15 ribbed Hydrangea.
 - 6. Staminodia present; petals more than 10 mm long Parnassia.
 - 6. Staminodia absent; petals less than 10 mm long 7.
- 7. Leaves ternately decompound Astilbe.
- 7. Leaves simple 8.
 - 8. Flowers solitary in leaf axils, or in short, leafy cymes 9.

- 8. Flowers in racemes or panicles 10.
- 9. Sepals 4; carpels 2 Chrysosplenium.
- 9. Sepals 5; carpels 3 Lepuropetalon.
 - 10. Petals pinnatifid or fringed; stem leaves opposite Mitella.
 - 10. Petals not pinnatifid or fringed; stem leaves alternate or absent 11.
- 11. Ovary 1-celled 12.
- 11. Ovary 2-celled 13.
 - 12. Inflorescence paniculate; stamens 5 Heuchera.
 - 12. Inflorescence racemose; stamens 10 Tiarella.
- 13. Stamens 5; leaves palmately lobed Boykinia.
- 13. Stamens 10; leaves not palmately lobed Saxifraga

PLANT IDENTIFICATION EXERCISE

1. Identification of an unknown. Select an unknown specimen and identify it by keying in an appropriate manual, flora, or monograph. Verify your results by reading a description, by comparing with an illustration or by checking with your instructor.
2. Preparation of a comparison chart. Select 5 or more specimens from the group provided by your instructor. Identify each by keying. Verify your results. Prepare a description of each similar to those in a flora or manual. Be sure characters and character states are in the same order. Select contrasting character states and prepare a comparison chart (see Figure 25-3).
3. Construction of keys. Construct a dichotomous key to these specimens using the information in the comparison chart.

COMPARISON CHART						
	Decumaria	Itea	Ribes	Parnassia	Heuchera	Saxifraga
Habit	Woody vine	Shrub	Shrub	Herb	Herb	Herb
Leaf arrangement	Opposite	Alternate	Alternate or on spur roots	Basal (Rosulate)	Basal (Rosulate)	Basal (Rosulate)
Petal Number	7-10	5	5	5	5	5
Locule Number	7-10	2	1	1	1	2
Stamen Number	7+	5	5	5 (stamonodia 5)	5	10
Fruit Type	Capsule	Capsule	Berry	Capsule	Capsule	Capsule

Figure 25-3. A comparison chart used in the construction of keys (for six of the genera in Figures 25-1 and 25-2).

Section B. RECENT AND NEW IDENTIFICATION METHODS*

**Adapted from "Specimen Identification and Key Construction with Time-Sharing Computers" by Larry E. Morse (Harvard University, Cambridge, Massachusetts), in Taxon 29: 269-282 (1971), with extensive revisions by Mr. Morse. Used with permission.*

I. POLYCLAVES

Polyclaves of various kinds allow one to select the characteristics for use in identifying each specimen, taking his choices from some character set and repeating an elimination process until a tentative identification is made. A printed data table, chart, or matrix giving the status of various taxa for useful characteristics is readily used as a polyclave by listing the possible taxa on scratch paper and crossing out those which do not agree with the specimen's characters. Such data tables appear irregularly in the taxonomic literature, often for only the more difficult groups involved but occasionally for all the treated taxa, as done for medical bacteria by Cowan and Steel (1965). For large groups, the diagnostic tables are not only more powerful than the equivalent key, but also take less space to print. Lists of taxa having various characters were among the first nontabular polyclaves. These resemble the inverted files common in computerized information systems, where entries are listed according to their characteristics rather than characteristics by entries. Lists of taxa lacking specified features have also been produced; this modification expedites use as one may then jot down the possible taxa and rapidly cross off those differing from the specimen. Polyclaves are readily mechanized, as shown by the familiar edge-punched cards and the less familiar window keys, as well as various mechanical devices. The possibility of a computerized polyclave was noted by Sokal and Sneath (1966) and by Williams (1967); implementation is straightforward once suitable data formats have been devised. The computerized polyclave system we developed at Michigan State University (Morse, in press) uses a General Electric Mark II timesharing system, but has also been tested on several other computers. Other computerized polyclaves include those of Boughey, Bridges, and Ikeda (1968), Dybowski and Franklin (1968), Goodall (1968), and Walker et al. (1968). As yet no comparison of these many parallel approaches has been made.

If an identification method requires, in general, use of all the characters in some list, it is no longer a

polyclave, for no user options remain in selecting characters. Several such character-set methods have been devised, mostly statistical. Some computerized "keys" are also based on this model (e.g., Gyllenberg, 1965; Bogdanescu and Racotta, 1967; and Rypka, 1971). Although well intended, character-set algorithms seem to offer no advantages over polyclaves. However, pattern recognition methods, using no formal characters, might be employed effectively for identification in some cases. Such techniques linked with optical scanners, spectroscopy, or chromatography could even offer fully automated identification.

II. TAXA, CHARACTERS, AND DATA MATRICES

An objective identification system requires a priori information of three kinds: the pertinent taxa, the useful differentiating characters of these taxa, and the taxon/character data themselves. ,

The hierarchy of natural taxa is the foundation of biological information retrieval as well as systematic synthesis, since data can be stored, retrieved, and studied at any level of generalization (Morse, 1974). In data processing, we may usually regard a taxon as a group of one or more individuals or lower taxa judged sufficiently similar to each other to be treated together formally as a single evolutionary or informational unit at a particular level in the taxonomic hierarchy, and sufficiently different from other such groups of the same rank to be treated separately from them. Traditionally each taxon (except the highest) belongs to one and only one taxon of the next higher rank, implying each individual belongs to exactly one species (and has one name) in any particular taxonomic treatment of its group. Taxa are either monothetic or polythetic. For monothetic taxa, possession of a certain set of diagnostic characters is both necessary and sufficient for membership. Polythetic taxa are more loosely circumscribed, since presence of only "a large number" (rather than all) of a list of characters is required for membership. In other words, the members of a polythetic taxon exhibit overall similarity. Modern systematists employ the polythetic taxon concept in most of their work, as has been widely recognized in the past decade. However, dichotomous keys and other monothetic methods of identification are still used. Theoretically, identification schemes for modern biology ought to assure that no single-character difference, either in population variability or observer error, can result in a misidentification. Polythetic polyclaves offer a solution; here, no possibility is eliminated until several differences have accumulated between the taxon description and the unknown specimen. The appropriate threshold depends on the variability of the taxa, but toleration of just one or two differences often gives a marked improvement in identification success.

Our character model was originally developed for a list-structure representation of dichotomous keys (Morse, Beaman, and Shetler, 1968). It centers on the concept of character couplets (or two-state characters), and also allows coding of "dependent characters" (Williams, 1969). In our model, each character couplet represents two contrasting alternatives about possible features of an individual population. (For convenience we often label the possibilities A and B, or true and false.) In other words, the character couplets are equivalent to the pairs (couplets) of contrasting leads in dichotomous keys, specifying a part of the organism and asking which of two alternative modifications of it is present. Such a question implies the specified part itself is present--the couplet "petals red" versus "petals yellow" is meaningless for apetalous plants as well as those having petals of different colors. A couplet can thus be inapplicable or non-comparable, often coded NC in numerical-taxonomy data. An additional code is needed when a specimen is variable in its expression of the character states of a couplet; more precisely, variable means sometimes A and sometimes B. At other times the state of a couplet is not known for a specimen, perhaps because the character is difficult to determine or not recorded, or the specimen is incomplete, poorly preserved, or at the wrong stage of development; these cases are coded unknown. All binary-couplet characters can be readily encoded with these five character states, listed below with our code numbers for them:

0 = unknown

3 = false, or B

1 = true, or A

4 =
inapplicable

2 = variable

Each character couplet in our model is a hierarchy of two binary (two-state) characters, one implicit and one explicit. Expression of either state of the explicit character depends on the presence of one particular state of the implied more general character--petal color depends on petal presence. As well illustrated in a diagram by Dale (1968), larger character hierarchies can be built by combination of a number of such couplets.

Extensions to multi-state and quantitative characters are, of course, necessary before our methods can be seriously considered for handling taxonomic information in general. For this we implemented a file system involving three kinds of taxon/character data: two-state characters, as described above; multi-state characters, and quantitative (numeric) characters (Morse, Peters, and Hamel, 1971).

Multi-state and quantitative characters may also be coded as couplets for use with the current programs. The former can always be expressed as a list of yes/no binary characters, while the ranges of quantitative characters may be segmented as appropriate. Details of the MSU programs are available elsewhere (Morse, in press).

For identification and allied procedures, the preparation of data matrices as taxon/character rather than specimen/character entries is preferable. In compiling effective taxon/character descriptions it is of course important to study an adequate sampling of specimens and other information, including the relevant literature, as otherwise the data cannot indicate the nature of the taxon as a whole. Yet if enough examples of a taxon were examined, virtually all the characters studied would eventually be marked variable, rendering the whole description useless. However, with polythetic identification methods the occasional oddity can be safely ignored in preparing a taxon description. For polythetic taxa, the code variable means noticeably variable, while A (or B) means usually A (or B), perhaps 80% (Cowan and Steel, 1965) or 85% (Lapage et al., 1973), or a number calculated from the data (Moller, 1962). In our present system, taxon/character data are presented in the format illustrated elsewhere (Morse, in press; Shetler et al., 1971). Briefly, such files consist of a taxon list, a character list, and a coded taxon/character data matrix, with additional fields for various parameters used by the programs. Allowance is also made for extensive documentation.

Hierarchical matrices aid in processing of taxonomic data since different characters may be used for different subgroups of taxa. In a hierarchical taxonomic matrix system, each taxon-description line may refer to another complete matrix differentiating subordinate taxa, permitting programs to work down through the hierarchy much as one uses a key to families, then a key to genera, and then a key to species. This method also allows the user to query the data bank beginning at any level he desires--if he knows his specimen is a Rhododendron, he need not first verify that it is an angiosperm!

In preparing identification data, one should include only characters of potential value in making identifications. A number of computer algorithms are available for character-set selection, but subjective screening appears ordinarily adequate for choosing a manageable set of characters for further study. The traditional key characters of the group offer an initial list, but new diagnostic characters should be sought, especially in difficult groups. Guidelines for character selection, offered in a multitude of works, include the following: presence in the usual material to be identified, ease of observation and interpretation,

distinctness between the various states, independence from other characters used, and tolerance to environmental influences. Identification characters must not only be well expressed in nature but should also be clearly expressible verbally or pictorially. Occasionally one should describe what the user will think he is seeing, whether or not this be the actual case morphologically. The previous experience of the intended users should also be considered, and especially in educational works the characters might be easily learned and remembered in association with their taxa.

III. CONSTRUCTION OF IDENTIFICATION KEYS

The presentation of dichotomous keys is usually considered a mandatory part of scholarly taxonomic publication, yet until recent years few advances were made in key-constructing theory over the methods of Grew or Lamarck, or even Aristotle and Theophrastus (Voss, 1952). In his study of the theory of keys, Osborne (1963) concludes that the most efficient key is the one in which each dichotomy divides the remaining taxa symmetrically, or as nearly so as possible-, assuming the characters have equal probabilities of misinterpretation and the taxa are equally abundant. However, in reality some characters are far easier to use than others, and some taxa are keyed out more often.

The relative conveniences or ease of use of the various characters should be a major consideration in key construction, as one wants the easiest and fastest key which is sufficiently accurate. Ledley and Lusted (1960) suggest the characters (disease symptoms in their paper) be grouped into numbered sets or blocks of similar difficulty. In making a diagnosis they employ all available characters from one block before using any from the next.. We allow nine such blocks, with the characters in block one hardest and those in block nine easiest; any characters labelled zero are completely ignored by some routines. In our programs we treat the block codes, or character convenience values as natural logarithms of a convenience-of-use-number. Thus the nine symbols represent a range of about three to eight thousand. When rating characters, one should assign codes such that the intended users would prefer to study any three characters at some level than any single character at the next lower level, seven to one for a two-level difference, twenty to one for three levels, and so forth. The conveniences may, of course, be revised with experience or a change in user groups. Hall (1970) considers conveniences which vary from taxon to taxon, but finds it easier to rate each character only once, instead of once per taxon.

If the relative frequencies of the various taxa can be estimated for a situation, then the taxa can be considered accordingly in determining the evenness of division at a dichotomy, in effect taking each expected occurrence as a separate object to be identified. Determining such frequencies is not at all easy, since we are interested in how often a taxon will be collected and keyed out, a quite different value than how often it occurs in the field.

The intra-taxon variability of the characters is also an important consideration in keys. In our methods, characters are coded as -either variables or as (nearly) constant, leaving selection of a variability threshold to the researcher preparing the data. An alternative approach often used involves stating the variability of each taxon for each character as a quantitative entry in the data matrix, usually as a percentage. This percentage matrix would be valuable in a general taxonomic information system, especially while data were being collected and variability patterns were developing. However, the true-false-variable matrices present their data more clearly, and take far less effort to prepare. Also, we know no programs which utilize the percentages directly in constructing keys.

Many texts recommend the use of data tables in writing keys, but other mechanical aids are rarely described. Metcalf (1954) presents an index-card technique, and Peters (1969) used a computer to help

incorporate additional taxa in keys. While developing our key-constructing techniques we prepared a key-editing program (Morse, Beaman, and Shetler, 1968), but work on the advanced system" outlined there was suspended in favor of our current research on key construction and identification procedures. Key editing is important in large projects such as Flora Europaea and the planned Flora North America, where numerous minor improvements are made in the keys during editing. However, if taxa are added or deleted, it is often better to develop an entirely new key.

The possibility of computerized key construction is often mentioned, yet we know of only four programs for construction of biological keys, namely those of Moller (1962), Hall (1970), and Pankhurst (1970, 1971), as well as our own (Morse, 1971 and in press). Pankhurst (1974) provides a comparison of some of these programs. Moller's method requires complete binary data, and has attracted little attention. Hall's program utilizes quantitative data, printing a numeric version of the key which must then be rewritten before use. Pankhurst's algorithm differs from ours primarily in his use of rigid character-convenience blocks and his employment of the attribute value rather than hierarchical-couplet character concept. His program, like ours, prints the key directly. The production version of our MSU programs is described below; these allow mixed-data key construction with our new data matrices. Several matrix-reduction and monothetic-deviative algorithms in the literature resemble key construction: the KEYCALC program by Niemalg, Hopkins, and Quadling (1968) is typical. Also, some aspects of decision-tree and game-tree research in computer science could contribute to the theory of keys.

In preparing a key, one usually divides the initial group of taxa by a character couplet into two subgroups, each of which is independently divided into further subgroups, and so forth, until every taxon is distinguished from all others. Indeed, the further subdivision of a subgroup can be considered as construction of a full key to that local group, suggesting a concise recursive algorithm for computerization: construct the key by dividing the original taxa into two subgroups by the best possible character couplet, then consider each of these subgroups separately, dividing them similarly. It is readily established that this procedure will produce a key if one can be made at all, but the optimality of such a key apparently remains uninvestigated. The four key-construction programs cited above all operate on this principle.

For use of this algorithm one needs a means for selecting the "best" character couplet for a given dichotomy, given the appropriate taxon/character data for a set of potentially useful characters. In our program, a preliminary screening of the characters is followed by a detailed evaluation of the potentially useful ones. In the preliminary scan, a character is eliminated from further consideration if it is coded as unknown or as inapplicable for any taxon in the local group, or if it is coded in only one way within that group. In other words, the remaining characters are all coded partly true and partly false; true and variable; false and variable; or true, false, and variable.

For keys these are obviously the only characters of immediate interest. Next, a combined measure of ease of use and dichotomizing power is determined for each character. A tally is made for each state (true, false, and variable), giving the sums of the taxon frequencies of the taxa so coded for a character. In effect, these three tallies give the total number of encounters expected under each state of a character. The tally for variable is then multiplied by the sum of the other two tallies, and half this produce is added to twice the product of the true and false tallies, giving a "dichotomizing value." $DV = 2qTqF + (1/2)qV(qT + qF)$,

where qT , qF , and qV are the respective tallies. High values of DV indicate well-divided dichotomies. In considering the character conveniences, these ratings are treated in exponential (values 3 through 8103) rather than condensed (1-9) form. The "best" couplet is found by maximizing the product of the character

convenience exponential and the dichotomizing value. The character couplet having this product greatest is then used for that dichotomy of the key, and the taxa in the local group are sorted according to their recorded status for that character, placing variable taxa in both subgroups. Note the character conveniences have a powerful but not exclusive effect, as a 5-5 split at convenience 6 (7254) as well as a 6-4 (or worse) split at 5 (7004).

If the indented style is desired, the program prints the key . as it selects the dichotomies. However, for the bracketed style the key must be stored (in our list-structure representation) until completed, since division of variable taxa affects the numbering. It is then printed in one operation. The list-structure condensation can also be printed if desired, perhaps for later use in editing.

Ordinarily a batch- processing computer is adequate for key construction since no user interaction during processing is necessary. However, on-line key construction allows the effects of changes in the data to be seen quickly. An on-line key-editing system would also be possible, but most changes can be made with the text-editing software usually supplied with time-sharing systems. For small groups, these methods may also be used manually..

IV. DYNAMIC POLYCLAVES

Polyclaves need not be automated, but might be computerized when a large, rapidly developing information base is involved; otherwise, imaginative publication methods suffice (e.g., Ogden, 1953; Leenhouts, 1966, Archbald, 1967; Duke, 1969; Hansen and Rahn, 1969; Shultz, 1973). Here on-line (usually timesharing) computers offer a clear advantage over batch-processing systems, since the on-line system allows the user to submit additional data during the execution of a program. This permits a dialogue or conversation between the user and the computer, with the machine printing questions and awaiting responses before continuing the processing. Such conversations can be about as terse or verbose as desired; we have taken a middle course, writing out most questions and taxon names in full, but sometimes using numerical codes for character states and couplets, as well as for several program options. Numeric coding of lengthy answers saves printing or typing time and reduces the chances for errors, assuming the user can copy a short number more accurately than a long descriptive phrase. For regular users of the system a faster abbreviated terminology is planned. As yet, the ideal of free-form language input is not practical for this or almost any information-retrieval system (Simmons, 1970).

Following selection of a group of possible taxa by the choice of the initial data file, our polyclave algorithm consists of three steps, repeated as necessary: (1) request the user to give one or more characteristics of his specimen; (2) eliminate all possibilities inconsistent with this partial description; and (3) print the results of the elimination, either an identification or some other action, and recycle to the first step if necessary. Several alternatives are available, including production of a useful-characters list and deletion of the effects of the last character set submitted. Since a user may start with a data matrix at any level in the taxonomic hierarchy, continuation to a subsequent matrix may be possible-after a family, genus, or other higher-level taxon is identified. After each identification, several diagnostic or peculiar characters of the taxon are listed as an immediate check of the suggested identification. Particularly with higher level matrices, "false positives" may occasionally occur when no single species has a character combination suggested by the generalized description of a taxon.

Polythetic identification, where one difference no longer implies elimination, is available as an option on our computer system. Here the program tallies the number of differences and eliminates a taxon only when its tally exceeds a user-determined value, commonly one, two, or three. Although it allows for greater taxon variability or user error, the polythetic polyclave is slower than the monothetic one since more characters must be submitted to assure complete elimination of all the other taxa.

When a list of useful characteristics is desired part way through a polyclave procedure, a subroutine selects the best characters to continue dividing the set of remaining possibilities. Actually, a portion of the key-constructing algorithm is used to determine the first such character, and repetition of the procedure (ignoring this character) gives the next-best character, and so forth. With the polyclave, of course, such recommended characters are merely suggestions which need not be employed. No matter which state the specimen shows, the recommended character will eliminate about half the possibilities. Any other character, on the average, will delete fewer taxa because it eliminates a larger number only when in its rarer state. However, if the user happens to notice his specimen displays a rare character, he can eliminate a large number of possibilities at once and identify his specimen much more rapidly. Since the ability to recognize rare characters and realize their power comes only through training and experience, the expert delights in the efficiency and power of a polyclave, while the neophyte is lost in its multitude of choices and prefers the supervision and security of the traditional dichotomous key.

Note: The bracketed key above shows that the two leads of the couplet are together. In the indented key below the possibilities of the first lead is worked on first before proceeding to the next lead. Notice that both keys are constructed using the same characters and character states.

E-tivity 8.2.2 Use of identification Keys

Numbering, pacing and sequencing	8.2.2
Title	

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Purpose	To describe the use and types of keys
Brief summary of overall task	Read: Dichotomous, bracketed and indented keys

Spark	
Individual task	Outline the points you should consider while constructing a dichotomous key. Do this in the discussion forum
Interaction begins	Comment on the answers given by your colleagues in the discussion forum
E-moderator interventions	1. Summarize the threads and review the scanned forms 2. Give feedback 3. Provide teaching points 4. Close the E-tivity
Schedule and time	This task will take 1 hour
Next	Classification of cultivated plants

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8.2.2 Classification of cultivated plants

The classification of cultivated plants is governed by the International Code of Nomenclature for Cultivated Plants, (ICNCP). The rules are taken from the relevant ones in International

Code for wild plants but below the species level there is a provision for recognition of only one rank, the cultivar. A cultivar is defined as an assemblage of cultivated plants which are clearly distinguishable by any characters (morphological, cytological, chemical etc.) and when reproduced sexually or asexually retains the distinguishing characters. To produce a satisfactory identification and classification system for cultivated plants it is necessary to have the following:

- Carefully prepared description
- Accurate illustrations
- Herbarium with cultivated plant specimens
- A basic classification into groups
- Publications of taxa

Cultivar Descriptions there is inadequate descriptions for most cultivars to works of high quality. This is unlike wild plants where there are good descriptions as in Floras such as the Flora of Tropical East Africa

Accurate illustrations

Illustrations of individual cultivars within most species are not adequate. A few are well illustrated but for the majority there is nothing. Wild plants are comparatively better.

Herbarium and living reference collection

Herbarium specimens are useful for identification and research especially for construction of classification systems. In herbaria cultivated plants are poorly represented usually by a few specimens. Ideally there is need for a living reference in form of a botanical garden for:

- Identification
- To act as gene bank so that genetic variation of the collections will be available for future breeding work and research.
- To act as sources of propagating material especially of rare and interesting plants, which can be propagated, distributed.
- For demonstration purposes

Reference living collections are found in specialized plant societies for special interests not for botanical purposes. Such societies include: local municipal authorities, educational establishments, and occasionally private individuals.

Basic classification into groups

Classification of cultivars into groups and systems of identification are often inadequate and poorly represented. There is need to describe the characters of groupings that have been made for ease of identification

Publication in a clear form

Publications have not been done well. They should contain a historical survey of the wild relatives, their introduction to cultivation and evolutionary trends.

Nomenclature of cultivated plants.

In 1862 Alphonse de Candolle suggested reserving Latin names for plants of wild origin and non Latin names for cultivated plants. The proposed rules were formally accepted at the 10th Horticultural congress held in London in 1952. The code was called “The International Code of Nomenclature for Cultivated Plants” (ICNCP). The code deals with plants that have been selected and maintained in cultivation by man and includes Plants of Agriculture, Horticulture and Silviculture.

One of the rules is that from 1st Jan. 1959 cultivar names should not be in Latin form but should be in modern language such as English using the Roman alphabet. Cultivar names should have an initial capital letter and either preceding the cultivar name by abbreviation cv. or be enclosed in single quotes e.g. cv Violet Flame, cv Crimson Glory. The code aims at fixity and stabilizing cultivar names. The rules are applied to a large number of cultivated plants but confusion still exists in the identification and indiscriminate naming of cultivars. To reduce the confusion there is need to register cultivar names and establish reference collection for identification and ensuring that new introductions are distinct from existing ones.

E-tivity 8.2.3 Classification of cultivated plants

Numbering, and sequencing	8.2.3
Title	
Purpose	To describe the rules and classification of cultivated plants and the challenges faced with cultivated plants.
Brief summary of overall task	Watch video Rules on cultivated plants conditions for new names
Spark	

Individual task	Describe the main challenges faced in the identification and classification of cultivated plants compared to wild plants. Post this in the discussion forum
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Interaction begins	Comment on the answers give by your colleagues in the discussion forum
E-moderator interventions	1. Summarize the threads and review the scanned forms 2. Give feedback 3. Provide teaching points 4. Close the E-tivity
Schedule and time	This task will take 1 hour
Next	Herbarium

8.3 Assessment questions

1. Which of the following statements does not describe a method for plant identification?

- a) Use of dichotomous keys
- b) Use herbarium specimens
- c) Use evolution of plants
- c) Use distinguishing characters

2. In an indented key all the possibilities that depend on the first lead of a couplet are dealt with before moving to the next lead T / F

3. The two leads of a couplet must always contradict T / F

4. Cultivated plant names are Latin or Latinized T / F

8.4 References

1. <https://www.biologydiscussion.com/plant-taxonomy/plant-identification/plant-identification-characters-and-methods/30279> 2.

Dichotomous and bracketed and indented keys <http://www.ibiblio.org/unc-biology/herbarium/courses/CHPT25.html>

3. Cultivated plants rules

<https://www.youtube.com/watch?v=i1wbkDuelcQ>

4. Conditions for new names <https://www.youtube.com/watch?v=i1wbkDuelcQ> 63

LESSON 9: HERBARIUM**9.1. Introduction**

In this lesson we look at one of the most important tool of a plant taxonomist which is the herbarium. The term herbarium was for the first time used by Joseph Pitton De Tournefort in 1700 to describe a collection of dried plants and since then it has been used in the same context. We can here define a herbarium as a collection of plant specimens, which have been pressed, dried mounted, identified and classified according to an approved system of classification. The term is sometimes used to refer to the building where such a collection is housed.

8.2. Lesson learning outcomes:

At the end of the lesson the learner will be able to:

8.2.1 Describe the different types of Herbaria

8.2.2 Describe the different types of herbarium collections

8.2.3 Explain the preparation and preservation of herbarium specimens

8.2.4 Discuss the functions of the herbarium, library and botanical gardens

9.2.1 Types of herbaria;

There are different types of herbaria depending on the size and the functions carried out.

International herbaria are major herbaria housing plant collections from all over the world. Examples are Paris herbarium with about 9m specimens, and the Royal Botanic Gardens, Kew, England with about 6m. specimens,

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National herbaria concentrate on plant specimens within a certain country. Under this is our East African Herbarium, Nairobi which has about 1m. specimens. Pretoria in South Africa belongs to this category and has about 1.2m specimens.

Local herbaria keep specimens within a small area such as a county.

Teaching or academic herbaria are found in teaching institutions such as universities where students train and deposit their collections. Examples are Nairobi and Kenyatta University Herbaria.

Private herbaria are kept by individuals such as the Linnaeus herbarium

Research herbaria represent specimens used for specific research such as on medicinal plants, weeds, particular families or specialized plant groups such as bryophytes.

E-tivity 9.2.1 Types of herbaria

Numbering, pacing and sequencing	9.2.1
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Title	
Purpose	To describe different types of herbaria
Brief summary of overall task	Read : types of herbaria
Spark	
Individual task	Outline the different types of herbaria and the criteria used for categorization. Do this in the discussion forum
Interaction begins	Comment on the answers give by your colleagues in the discussion forum

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E-moderator interventions	1. Summarize the threads and review the scanned forms 2. Give feedback 3. Provide teaching points 5. Close the E-tivity
Schedule and time	This task will take 1 hour
Next	Types of herbarium collections

Kinds of Herbaria

The kind of herbaria depend on the purpose, region, contents, plant groups present in these:

1. Herbaria of medicinal plants

This type of herbaria includes the specimen of plants having medicinal importance/medicinal properties.

ACECR; Central herbarium of medicinal plants, halejerd region, Karaj, Alborz province.

2. Herbaria of weeds

These herbaria contain weeds of cultivated fields & waste places.

3. Regional herbaria

As the name indicates, herbaria of a particular region or place.

Example: Herbaria of the central circle, Allahabad.

9.2.2 Types of collections

In a herbarium there are many types of collections. There is the main collection made up of mounted specimens mostly of angiosperms and pteridophytes and there are ancillary collections which include fossils, fungi, lichens, bryophytes, seeds, fruits and wood. Sometimes there are also specimens stored in preservative fluids in specimen bottles. These ancillary collections are stored separately in the herbarium.

Main collection

The main collection, are the mounted specimens consisting of pteridophytes, angiosperms and sometimes gymnosperms.

Carpological collections

A carpological collection consists of cones, fruits and seeds each cross-referenced with corresponding herbarium sheet. They are kept in boxes or bottles in shallow drawers. .

Xylarium

This is a collection of wood specimens for anatomical studies or for identification purposes. Wood blocks can be stored in cupboards or drawers. Voucher specimens are kept in the main collections with the catalogue number of the wood sample

Bryophytes and Lichens

These are kept in folded envelopes.

Macrofungi.

Stored whole or sliced lengthwise. Fungi are susceptible to insect attack and should be disinfected thoroughly before storage and kept dry with silica gel.

Fossils

Fossils are usually large and bulky and are kept in metal filling cabinets with shallow drawers.

Spirit collection

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Delicate flowers, whose features are important for identification such as Orchid flowers, are kept in spirit collections (in preservative fluid for keeping specimens from microbial attack)

Illustrations and photographs

These provide data on the habit, habitat and colour of the plant. Prints and slides can be used in publications and lectures.

Photographs, Photocopies or scanned images may also record specimens from other herbaria such as types, and historical specimens.

Microscope slides

Slides with pollen, anatomical sections or other preparations are kept in special cupboards with drawers made to hold them.

E-tivity 9.2.2 Types of herbarium collections

What is a herbarium?

A herbarium is a collection of preserved plant or fungal specimens. Many Herbaria were established throughout Europe in the 1600s, during the age of exploration when botanical gardens could no longer keep living collections of all the known species and thus preserving and storing specimens became common practice.

The largest herbarium in the world is at the National Natural History Museum in Paris, France, with over 9.5 million specimens. The largest herbarium in the United States is the New York Botanic Gardens Herbarium, established 1891, with over 7 million specimens.

Many Herbaria in the USA were established throughout the 1800s. The USA has 405 registered herbaria with a total of 59 million specimens or 22% of the world's herbarium collections.

Herbarium specimens include plants, conifers, ferns, mosses, liverworts and algae as well as fungi and lichens. Most plant specimens are dried by pressing whereas bulkier plants and most fungi are dried without pressing and are stored in boxes.

Pressed specimens can be mounted on archival sheets or stored in packets, as in the case for most of the material in the Arthur Fungarium, for example.

Additional names can be used for more specialized collections, for example:

- A collection of fungi is a Fungarium.
- A collection of plants is a Herbarium.
- A collection of wood specimens is a Xylarium.
- A collection of cultivated plants is a Hortorium.

What is a type specimen?

According to a precise set of rules laid down by the **International Code of Nomenclature** for algae, fungi and plants (ICN), the scientific name of every taxon is almost always based on one particular specimen. This is usually a physical specimen that must be accessioned in a museum or herbarium research collection.

Types are of great significance. When identifying material, a researcher attempts to apply a taxon name to a specimen or group of specimens and this will be based on their understanding of the relevant taxa, using (at least) the type description, but preferably based on an examination of all the type material of all of the relevant taxa. If the taxon appears not to have been named, then the scientist or another expert designates a type specimen and publishes a new name and an official description.

This process is crucial to the science of biological taxonomy and biology as a whole and these type collections are considered the lynchpin of biological research.

Types of types

There are actually several types of types. Here are the definitions of some common types:

- **Holotype:** When a single specimen is clearly designated in the original description, this specimen is known as the holotype of that species.
- **Isotype:** Any duplicate specimen of the holotype.
- **Lectotype:** Designated as the type when no holotype was identified by the original author, or when the holotype is lost or destroyed. It is chosen from among the specimens available to the original publishing author.
- **Syntype:** Any of two or more specimens listed in the original description of a taxon when a holotype was not designated.
- **Isosyntype:** A duplicate of a syntype.
- **Paratype:** A specimen not formally designated as a type but cited along with the type collection in the original description of a taxon.
- **Neotype:** A specimen chosen by a later researcher to serve in place of a holotype when all specimens available to the original publishing author of a scientific name have been lost or destroyed.
- **Topotype:** A specimen of a plant collected from the same locality as the holotype and usually on a different date. A topotype has no formal standing.
- **Cotype:** A term formerly used for syntype and sometimes (erroneously) for

isotype and paratype. This is an old term that was used loosely and is not used by today's taxonomists.

What can you do with a herbarium collection?

Basic functions and Research

- Facilitate the discovery or confirmation of new species and provides ecological and range data for existing species;
- Voucher scientific research;
- Repository for type specimens;
- Provide information on rare or extinct species;
- Material for morphological studies and documentation of flowering, and fruiting times;
- Species locality data;
- Provide data for agriculture, human health, biosecurity, forensics, control of invasive species, conservation biology, natural resources, and land management;
- DNA analysis and sequencing. (For example, using herbarium specimens of potato and tomato leaves infected with *Phytophthora infestans* for the first time scientists **have decoded the genome of a plant pathogen and its plant host** from dried herbarium samples);
- Document specialist's concepts;
- Provide information on predators, pathogens and symbionts that may have been collected with the specimen (For example, the larvae of some *Mycodiplosis* species feed primarily on the spores of rust fungi and **studies on these larvae** have used herbarium specimens dating back to the 1880s);
- Document CO₂ changes and minor cycles in climate; • Information on expeditions and explorers.

Education and training

- Material for teaching;
- Research projects;
- Run public workshops and awareness programs;
- Internships and job opportunities;
- Specimens for museum and educational exhibits. •

Numbering, pacing and sequencing	9.2.2
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Title	
Purpose	Describe the different types of herbarium collections
Brief summary of overall task	Read: Types of collections
Spark	
Individual task	Outline the different types of ancillary collections in the herbarium. Answer in the discussion forum
Interaction begins	Comment on the answers given by your colleagues in the discussion herbarium
E-moderator interventions	1. Summarize the threads 2. Give feedback

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	3. Provide teaching points 4. Close the E-tivity
Schedule and time	This task will take a minimum of 1 hour
Next	Preparation and preservation of herbarium specimen

9.2.3 Collection, preparation and preservation of herbarium specimens

Plant collecting

Taxonomists depend a lot on herbarium specimens for constructing classification systems, identification and research. It is important therefore to have large numbers of specimens and of high quality.

What to collect

What one collects depend on the purpose for which the collection is made. For research work the collector collects plants related to the research project, usually confined to plants of a particular family, genus or species. For herbarium specimens, collections could be made in poorly collected areas. Herbarium specimens should show range of variation and a detailed picture of the distribution of the species. Further collections are made for those species where these conditions are not met.

Where possible the specimen should have flowers and fruits, as these are in most cases important for identification.

The specimen collected should be large enough to fill the herbarium sheet. If plants are small select a number of plants to show variation within the population.

If the plant is small collect the whole plant including the underground roots. Basal parts of grasses, sedges, ferns and bulbous plants should be collected as these are essential for identification.

For trees and shrubs, representative specimens with flowers and fruits should be collected. Collect a number of duplicates. Duplicates provide material for exchange or gifts to other herbaria and ensure that if the specimens are destroyed or lost in any way there will be specimens left.

Field Notes:

The information recorded in field notes is as important as the plant specimen. In the field notes, all the information about the plant which cannot be observed from the dried specimen but could be useful for identification or research is recorded. Herbaria have collecting labels but if they are not available it is possible to make some. All notes should be made when the plant is being collected and not left to a later time, as it is possible to forget some details.

The following data should be included:

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Flora of: country and province.

Name: scientific.

Vernacular name: name and language in brackets

Locality: precise locality and should include a locality shown on an easily available map. Start with the larger area and descend to the smallest area e.g. If a plant was collected in KU, the locality should be written, Nairobi, Kenyatta University, 20km on the Nairobi-Thika highway. Some collectors mark the locality on a small map that is attached to the herbarium sheet. Exact localities are important to researchers who would want to go to the place to collect another specimen also for locating distribution. **Altitude** in meters above sea level Latitude and longitude.

Habitat: Indicate the ecological aspects of the collection e.g. the type of vegetation, the dominant species, type of landscape, whether on a slope, mountain, swamp etc. Type of soil, sandy, clay, roam etc.

Description: these include all the features that are not seen on the specimen but which would be valuable especially for identification. Examples of such information include, habit (tree, shrub or herb) height, diameter of stem if a tree, if a herb, annual or perennial, erect or prostrate, variation in size among others of the same species, whether the plant grows singly or in groups, colour of the flowers and fruits, smell, milky latex, time of opening and closing of flowers, frequency of distribution, observed pollinators. **Uses,** if medicinal, the part of the plant used.

Collector name and initials

Collector's number: these should be serially, starting with number one and continuing without repeating a number so that reference to the plant can be known by the collector and number. All duplicates must have the same number as the original specimen. **Date:** the date of collection (date, month and year), this is important as it indicates the time when one can expect to get the plant in flower in case of a need for further collections.

The plant press

When plants have been collected, they are directly put into press or they can be put in plastic bags, which prevent the specimens from wilting. Each plant should be put in its small plastic bag and the small bags are then put in a large bag.

The purpose of pressing is to flatten, dry the specimens, and display them in a way to show the different parts easily. The flowers are arranged in a way that they are not crumbled and some are opened to show the inside. The upper and lower surfaces of the leaves are well displayed. Specimens too large for the mounting sheet are bent or broken when still fresh.

The plant press consists of the following:

Two rigid metal or wooden frames slightly larger than the mounting sheet

There are two strong straps to tie the press

The specimens are arranged and displayed in newspapers, Each collection in its own newspaper.

Drying paper separates the newspapers with the specimen.

If available corrugated cardboard sheets are used as ventilators or spacers to facilitate in drying the specimen.

Steps used in arranging the press:

The frame

Cardboard

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Two sheets of drying paper.

Newspaper with the specimen

Two drying papers followed by another newspaper with a specimen etc.

Use a cardboard ventilator after every five to eight specimens

Finally a cardboard ventilator and the second wooden frame and then tighten with the straps.

Special pressing methods for succulent plants:.

Succulent plants have a lot of water and tend to continue growing in the press. If they are pressed with the ordinary specimens they make drying difficult and the specimens end up becoming moldy and useless. The following methods can be used:

- Plants can be cut longitudinally, and the fleshy inner parts scrapped out before pressing (the method works well for *Aloe* species).
- The plants can be killed by submerging them in boiling water or in methylated spirit, petrol or household vinegar.
- The plants can be kept in a freezer for two days and then microwaved before pressing.

The press is put in the sun to dry or sometimes artificial heat is used. However care should be taken not to overheat the specimens, as this would make them brittle. Drying papers should be changed daily until the specimens are completely dry.

MOUNTING

The main advantages of mounting specimens are:

- To make the specimen easier to handle and more accessible for study and reference.
- To display the specimen to allow maximum observation of most characters
- To prevent damage and to preserve the specimens by attaching them to strong mounting sheet.

Specimens are mounted onto strong good quality paper of standard size (27cm by 42 cm). The specimens are attached to the mounting sheet by glue, White gummed paper or self adhesive paper cut into strips. Thread is used with a thick needle for bulky parts. Sandbags weighing about 5kg are used for putting pressure on a stack of specimens when glue is used. Small loose parts e.g. flowers or seeds are put in paper envelopes and attached to the sheet. The label with the field notes is put at the bottom right hand corner of the mounted sheet.

PRESERVATION OF HERBARIUM SPECIMENS

Insects and molds destroy herbarium specimens.

The pests most encountered in herbaria are cigarette beetles, cockroaches, rats, and fish moths and booklice.

Humidity is not a problem in most dry climates but in some areas in the tropics there is high humidity. The condition is conducive to fungal growth and insect production especially cockroaches, fish moths and booklice. It can be controlled by air conditioning at 50%. Another alternative is to put a desiccant such as silica gel in the herbarium cupboards. However the gel should be replaced every three months

Protection against Pests

- To protect against pest attacks various precautions are taken.

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- The specimens are put in special constructed cabinets with closely fitted doors to prevent cockroaches and rats from reaching the specimens.
- To prevent insect damage by incoming specimens, the specimens can be freezed at – 18 degrees centigrade.
- Incoming specimens can be microwaved but although this method is cheap, quick and safe it can denature the specimens by destroying the DNA and pollen. The other disadvantage is that only few specimens can be microwaved at a time
- The collection in the herbarium can be fumigated using volatile compounds such as methyl bromide which are effective against adult beetles and eggs. Fumigation is done

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by registered Pest Control Company. It is effective for about one year after which it is

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repeated.

Localized treatment: Individual specimens that are infected are sprayed directly with a dry insecticide. Non-volatile insecticide such as barium fluorosilicate, starch and glue mixture can control fish moth infestations when painted beneath the shelves or on the back of compartments of wooden herbarium cabinets.

Chemicals previously used such, as mothballs (naphthalene) which were put in herbarium cupboards are infective and toxic to humans. Mercuric chloride has been used for a long time but is very harmful to humans therefore is not recommended.

E-tivity 9.2.3: Preparation and preservation of herbarium specimens

Numbering, pacing and sequencing	9.2.3
Title	
Purpose	To prepare good quality herbarium specimens
Brief summary of overall task	Read collection of herbarium specimens

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Spark	
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Individual task	Collect 5 good quality herbarium specimens of different species and post the field notes in the discussion forum
Interaction begins	Comment on the notes taken by your colleagues in the discussion forum
E-moderator interventions	1. Summarize the threads and review the scanned forms 2. Give feedback 3. Provide teaching points 4. Close the E-tivity
Schedule and time	This task will need a minimum of 2 hour
Next	Functions of the herbarium

9.2.4 Functions of the herbarium

Identification. The basic work of a herbarium is to accurately identify plants and making this and other information on herbarium specimens available.

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Record of plant names

Training of botanists. Herbaria that are associated with universities provide graduate training in plant taxonomy.

Preservation of data on vegetation. Most plants occurring in different regions of the world are well represented in herbaria. Many plant species are lost through cutting down trees for various products, cultivation; overgrazing etc. the only documented evidence on some of the species is on the herbarium specimen and the notes accompanying the specimens.

Preservation and caring for valuable historical dried plant material. These are mainly type specimens.

Taxonomic research. Much of taxonomic research including construction of classification systems require large numbers of specimens and species often of many geographical regions and are based in herbaria such as Bentham and Hooker natural system and the Hutchinson phylogenetic system.

Writing of Floras such as “*The Flora of Tropical East Africa*” at the herbarium in Kew, England. Monographs and revisions are done at the herbaria.

Herbaria lend great numbers of specimens for taxonomic research to other institutions. Graduate students in the preparation of doctoral thesis use a very large proportion of these loans.

Herbarium specimens are also useful in determining the geographical and morphological limit of the species. No one person can hope to see in the field any large number of species and throughout their geographical range but in the herbarium.

Anatomical Research. The dried plants can be revived for use in anatomical studies.

Cytological Research. Some herbaria are associated with botanical gardens. The facilities of the gardens provide a fine opportunity of growing living materials for cytological work. These studies include the pairing of chromosomes at meiosis studies using young anthers and root tips for chromosome number and the herbarium provides a place where documented specimens of plants, which have been investigated, to be permanently deposited. Such specimens are called Voucher specimens.

Palynological Research. Pollen grains are little altered by the process of preparing specimens for the herbarium and as a result herbaria have proved useful as a source of palynological data.

Phytochemical research. Some chemical research can be done using herbarium specimens especially studies on flavonoid compounds.

BOTANICAL GARDENS

These are gardens which grow plants under ordinary conditions as well as controlled glasshouse conditions. There are special constructions for bryophytes and pteridophytes. In temperate climates there are greenhouses with controlled temperature and humidity conditions for tropical plants. Even in the tropics an indoor growing space is necessary except for the drought tolerant species.

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Uses of botanical gardens:

1. Aesthetic purposes. Many plants in the gardens are ornamental and are planted in organized manner.
2. Taxonomic research. A wide range of species can be grown for anatomical, cytological, chemical or breeding work.
3. Propagation of rare, threatened and interesting plants for study in relation to conservation and setting up of gene banks.
4. On site teaching. Collections of plants are often displayed in such a way that they can be used for self-instruction or demonstration purposes.
5. Most well organized botanical gardens operate an informal seed exchange scheme. Annual lists of available species are exchanged between botanical gardens. Small samples of seeds are exchanged free of charge.

LIBRARY

Established herbaria have libraries with publications of past botanical works. This is necessary because taxonomist rely heavily on accumulation of literature. For example tracing the past history of taxa to establish name changes, past descriptions, dates of publications and type specimens. Herbaria in developing countries lack extensive libraries and this limits the kinds of taxonomic work, which can be carried out.

E-tivity 9.2.4. Functions of herbarium

Numbering, pacing and sequencing	9.2.4
Title	
Purpose	To describe the functions of a herbarium
Brief summary of overall task	Read : herbarium techniques and functions botanical-gardens functions

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Spark	
Individual task	Outline the functions of an international herbarium. Do this in the discussion forum
Interaction begins	Comment on the answers given by your colleagues in the discussion forum

E-moderator interventions	1. Summarize the threads 2. Give feedback 3. Provide teaching points 4. Close the E-tivity
Schedule and time	This task will take 1 hour
Next	Reproductive data in plant taxonomy

.9. 3 Assessment questions

1. The East African Herbarium, Nairobi is an example of a:

- a) Regional herbarium
- b) Teaching herbarium
- c) International herbarium
- d) National herbarium

2. Xylarium is collection of;

- a) Fossils
- b) Wood specimens

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- c) Bryophytes and Lichens
- d) Microscope slid

3. Starch and glue mixture in the herbarium can be used to control:

- a) Fish moths
- b) Humidity
- c) Rats
- d) Cockroaches

4. The herbarium is important for taking care of type specimens T / F

9.4 References

1. Collection of specimens

<https://www.floridamuseum.ufl.edu/herbarium/voucher.htm>

2. Types of herbaria

https://www.globinmed.com/index.php?option=com_content&view=article&id=43:type_s-of-herbarium&catid=154&Itemid=130

3. Types of collections https://ag.purdue.edu/btny/Herbaria/Pages/What-is-an-Herbarium-and-what-does-it_do.aspx

4. Herbarium techniques and functions

<https://www.slideshare.net/BijuCherupuzha/herbarium-10212042>

5. Botanic gardens functions

https://www.biologydiscussion.com/plant-taxonomy/botanical-gardens/botanical_gardens-definition-functions-and-history/47602

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LESSON 10: REPRODUCTIVE DATA IN PLANT TAXONOMY

10.1 Introduction

The history of reproductive data in plant taxonomy as it relates to plant taxonomy begins with Rudolf Camerarius, a German who demonstrated the sexuality of plants in 1694. He proved that pollen is necessary for pollination and seed formation in the life history of plants. This has been followed by observations on the role of insects and other vectors in pollination.

10.2 Lesson learning outcomes

At the end of this lesson the learner will be able to:

10.2.1 Describe abiotic pollination mechanisms

10.2.2 Discuss biotic pollination mechanisms

10.2.3 Discuss other reproductive data: dispersal systems, phenology, ultraviolet patterns nectar and breeding systems,

10.2.1 Abiotic pollination mechanisms

Reproductive data can be used for comparative purposes or for understanding the biology of flower characters. As pointed out earlier taxonomists rely a lot on flower characters for constructing classification systems, identification and other uses and hence the importance of knowing the biology behind the characters.

Wind pollination (anemophily)

Wind pollination is very common and many families are primarily pollinated by it, for example Cyperaceae, Juncaceae and Poaceae.

Adaptations:

Lack of attractants such as attractive petals, fragrances and rewards like nectar and food bodies. To develop them would be a waste of energy and material.

The flowers are inconspicuous.

Sepals are reduced and petals are lacking.

Flowers have long flexible filaments which are easily swayed by the wind and the anthers are versatile.

The flowers produce abundant, dry, light, smooth and small pollen grains 20-30 μm in diameter to prevent the pollen from being too heavy to be airborne and to increase the dispersal units, as the probability of the pollen grain reaching the stigma is low. In *Pinus*

where the pollen grains are relatively large, 50-150 μm they are provided with wings to facilitate distribution by wind.

They have large feathery stigmas to increase the surface area for catching pollen. The flowers occur in large numbers to increase the total stigmatic area.

.ii) Water (hydrophily)

Water is a less common vector of pollination. It is found in some aquatic plants particularly the submerged species such as *Vallisneria* (Hydrocharitaceae) Adaptations:

Water pollinated flowers are small and inconspicuous, they are usually dioecious and have reduced perianth parts

E-tivity 10.2.1 Abiotic pollination mechanisms

Numbering, pacing and sequencing	10.2.1
Title	
Purpose	To describe pollination mechanisms by wind and water
Brief summary of overall task	Read: wind pollination . Video: wind pollination Water pollination

Spark	
Individual task	Outline adaptations to wind and water pollination mechanisms. Answer this in the discussion forum
Interaction begins	Comment on the answers by your colleagues in the discussion forum
E-moderator interventions	1. Summarize the threads 2. Give feedback 3. Provide teaching points 4. Close the E-tivity
Schedule and time	This task will take 1 hour

Next	Biotic pollination mechanisms
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10.2.2 Biotic pollination mechanisms**B) BIOTIC VECTORS**

These include insect pollination, bird and mammal pollination.

When insect –flower association began, about 120 million years ago, flowers and animals did not show a lot of adaptations to each other in terms of pollination. It is thought that Insects visited all kinds of flowers not recognizing the different plant species. Evolution has increased plants distinctiveness and ease in recognition by insect e.g. flower smell, size, shape, colour etc. Rewards like nectar, pollen and food bodies have also evolved. Insects have also evolved increased ability to recognize the most nutritious flowers. Thus flowers and insects have undergone coevolution. Coevolution has also occurred between flowers and bats and between flowers and birds.

Attractants and rewards to animal pollinators:

- a) Pollen grains are especially important for bees which take large amounts for their young
- b) Nectar is contained in nectary glands mostly situated at the base of one of the floral whorls.
- c) Scent: some flowers are attracted to sweet fragrances especially the nocturnal flowers e.g. *Cestrum* (Queen of the night), which has to produce powerful smells to direct insects to the flower. Some flowers are attracted to foul smells and are pollinated by carrion flies e.g. *Arum* flowers.
- d) Colour gives visual attraction to pollinators. Petals are the main advertisers of the flower but sometimes when they are small and inconspicuous they are supplemented by other parts, which are coloured and showy, for example the coloured bracts of *Bougainvillea*, *Euphorbia pulcherrima* (Poinsettia) and the red stamens of *Callistemon* (Bottle brush). Sometimes when flowers are small and inconspicuous they aggregate into inflorescences to form attraction units for the pollinators for example the compositae flowers,

i) INSECT POLLINATION (Entomophily)

Entomophilous flowers have sticky, ornamented pollen grains sometimes aggregated together such as the pollinia of orchids. Stigma is also sticky and pollen grains, nectar and food bodies are the main rewards. The following are some adaptations of flowers to specific insect pollinators.

Bee pollination (Mellitophily)

Bees are the chief flower pollinators. They visit more flowers than any other group of insects. Flowers are usually blue or yellow, day opening, mostly with a landing stage. The rewards are pollen and nectar. They visit chiefly the climax groups of angiosperms e.g. Lamiaceae, Papilionaceae and Compositae.

Fly pollination (Myophily)

